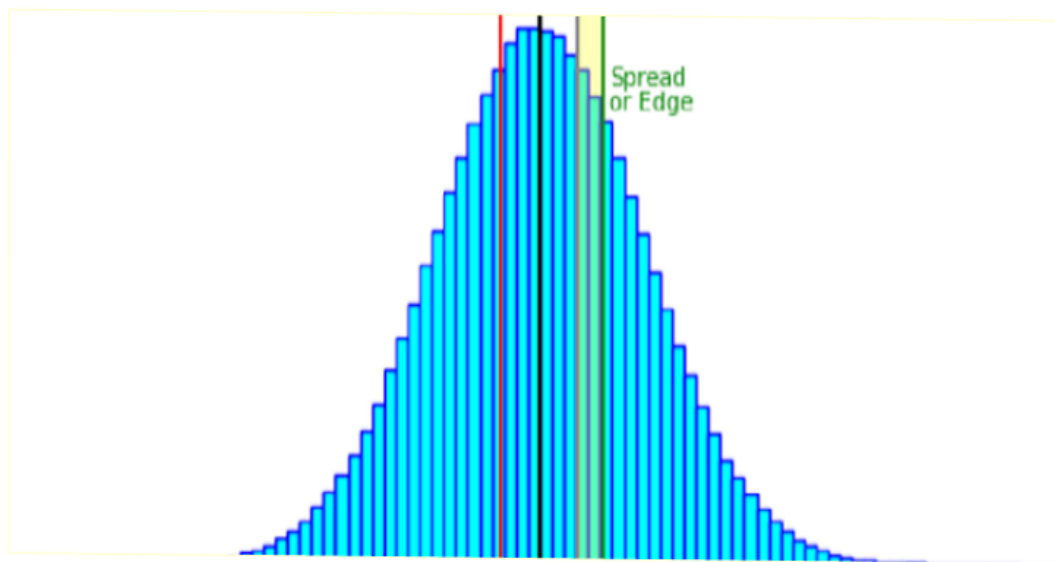


GAIN YOUR FINANCIAL FREEDOM



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Gain Your Financial Freedom

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Books Available On Amazon:

[Build The Retirement Fund You Deserve](#). *Amazon*, 2023.

[The Future Belongs To The Builders Of Mega Funds](#). *Amazon*, 2019.

[Reengineering Your Stock Portfolio](#). *Amazon*, 2019.

[Beyond the Efficient Frontier](#). *Amazon*, 2018.

[From Zero-Beta to Alpha Generation](#). *Amazon*, 2017.

[A Quest for Stock Profits](#). *Amazon*, 2017.

[Building Your Stock Portfolio](#). *Amazon*, 2017.

[Stock Trading Strategy Mechanics](#). *Amazon*, 2016.

[Trade Slicing Stocks](#). *Amazon*, 2016.

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Disclaimer: This book is not a solicitation to buy or sell securities. Past simulations are no guarantee of future performance. Presented simulations are based on actual past market data. What is said, however, should be considered just as common sense. All equal signs in all equations will prevail. As with most things, you have to decide for yourself. Understand the concepts and proposed trading methods, then extract what you need and adapt this content to your needs and preferences.

INTRODUCTION

We look at the future and want financial freedom and independence. We look around and can see the affluence of some and the poverty of others. We all would like to escape the latter group and join the affluent.

Our adult life revolves around work, family, friends, and the size of our paycheck. We usually live within what we can afford. Many try month after month to make ends meet until they retire. And then what?

We have to prepare. We have to plan for the long term. It is not just a wish. It is what we need to do. We cannot count on the government or anybody else to bail us out when the time comes, as it will come.

Understandably, once retired, you aspire to have enough money to do whatever you want. But you might have little time or capital to invest or start a business. Nevertheless, you need to prepare and do the best you can with what you have.

Gain Your Financial Freedom *is about your monetary future and what you can do now that will get you there. That is, reach financial independence beyond what you thought possible, preferably before you retire. If you like the methods, you could continue applying them while in retirement and continue to profit from them. They will only take a little of your time.*

PROLOGUE

I have these objectives in sight.

- Help you build a stock investment/retirement fund that remains under your control before and after you retire.
- Gives you the tools not only to build a substantial retirement fund but also to have it prosper even during retirement, where you could retrieve whatever you need for your living expenses and stuff.
- Give you the mathematical foundation explaining how you can do it and why it will all work.
- Show you how easily it can be done. Not only to reach a more than reasonable retirement income but also to push your investment fund to astonishing levels, should you want to.
- The last point: "should you want to" will be a recurring theme. Nothing presented gives you any obligations. You are free to choose what is best for you, whatever it is.

It will all come down to simple choices you can make. And then, living with what comes after those choices as if taking a fork in the road.

Will you do it or not? It will be your choice to make.

At least, you will know what was available and attainable. I intend to blow your mind with some of this stuff as we dig deeper and deeper, showing boundaries you might have thought unreachable.

I will deal primarily with logic, historical data, and mostly common sense stuff, things you already know or have not looked at in the same manner as they will be presented.

I will use century-old equations to make my points. Present new equations to illustrate better what was achieved and what could be achievable. These equations will make forward portfolio projections using the strategy's past behavior. The equations will also apply to your portfolio since they can be scaled to whatever level suits you best.

At all times, the equal sign will be the master of the show.

Moreover, you can verify and demonstrate that those equal signs will prevail. There is no better proof than the one **you** can ascertain. An equal sign can be extremely brutal; it does not offer maybes, opinions, or feelings. It is equal, or it is not. There is no middle ground.

In a trading ecosystem, I go for the practical side of things: **it works or it doesn't**.

Theories may be fine but often have little use in a quasi-randomly evolving future marketplace. Moreover, if theories do not corroborate or are impractical in the stock trading world, I put them aside and let them be.

What follows will fly in the face of Modern Portfolio Theory (MPT) and its Efficient Market Hypothesis (EMH).

I am not content with an optimal portfolio residing on the efficient frontier. You might have one of those over past market data, but that optimal portfolio is unattainable on that unknown future "efficient frontier".

A "what was" is no guarantee of what will be. Where will that efficient frontier be in 20 years or so? And what will be its composition? Can anyone tell? Why set such limits when there is much better available, feasible, and easily attainable?

Let what will follow also be part of your financial journey to go beyond what you thought possible, beyond MPT's market efficiency limits.

Jump over the efficient frontier where the grass is much greener¹. You will be able to go further with what is coming.

¹ I have a book on that on Amazon: [Beyond the Efficient Frontier](#). No need to buy it.

GAIN YOUR FINANCIAL FREEDOM

REACHING RETIREMENT

You will reach retirement age. The source of your primary income – your job – will end.

Regardless, you will still need money to enjoy that retirement.

Should you already have more than enough money to support all your retirement needs and wants, then mission accomplished. No need to read any further. Unless you are curious enough and still want to know more.

If you would like encouragement and new insight, read on. We will look at the investment/retirement fund thing from a different perspective and explore avenues you might have never considered.

Let's put some background into this.

Your savings account or pension plan might not be enough in your retirement planning. You cannot spend your retirement years on a rocking chair looking out the window.

You should want to be free and have the means to do whatever you want, enjoying life to the fullest with family and friends.

To get there, you should take some years before you retire to build that source of revenue for while in retirement, which may last more than thirty-plus years.

Most of your retirement income stream might come from what you have saved over the years: savings account, pension plan, property, investments, and other assets. However, you have to invest in assets that could bring higher returns to reach those higher income levels. Otherwise, you will have to make do with what you have accumulated over all those years of work.

The economic system is not designed to significantly improve your financial situation in retirement. Its primary function is to maintain your lifestyle based on the average income of your last few years of work. The higher your monthly income, the more you could save and invest in your future. However, as mentioned, savings alone might not be enough.

The key to enhancing your retirement income lies in your investment decisions.

It's not enough to hope for good returns. You need to actively seek those higher investment returns. The sooner you start, the better off you will be.

For instance, relying on a savings account with a 0.5% to 5% interest rate is not sufficient to fund the higher end of a comfortable retirement. To illustrate, apply the future value formula to your current savings.

At the 0.5% savings rate, with 3% inflation, you would effectively lose money over the long run. Even at the higher end of savings rates, say at 5% interest, your savings would only grow at the equivalent of 2% per year. It would take 35 years to double your money's buying power.

Starting at 30 and investing \$100k at that rate would give you a buying power of \$200k by the time you reach 65! Do you think it would be enough to retire on? After retiring, running out of funds may take only a few years. You have to protect yourself and your family and plan for a better future.

Hopefully, you have already realized that returns such as those given above are insufficient, especially when considering future inflation and the rising costs of living.

You have to explore more effective and more productive investment methods.

In short, **you need better solutions.**

In my article, [High Stock Portfolio Returns? Easy](#), I made what some might consider almost unbelievable claims. Not that much different from all the high-return hype you can see all over the web. Yet, my claims still hold. Moreover, anyone can verify any of them by redoing the same simulations I did, getting the same results, and reaching the same conclusions².

There is nothing hidden or secret sauce in all this, as I will demonstrate the worthiness of those claims. You can verify them yourself using the same code that generated what will follow.

The link to the original program code and its trading rules were also provided in the first and second articles of the nine-part series: **One Percent Per Week** trading strategy³. So, should those claims remain unbelievable or become more acceptable once you verify them? It is entirely up to you.

Anyone in his or her right mind with the required initial capital would have loved to invest in Renaissance's Medallion Fund, which, over the last 30⁺ years, before fees and expenses, has a reported CAGR of $\approx 63\%$. But this came with higher fees and

² The link to the strategy code is provided in the related files section.

³ Explore the series in related files at the end, especially the more recent.

expenses (4/44), compared to (2/20) for the average hedge fund. After those fees and expenses, the fund's net CAGR drops to $\approx 39\%$. Notwithstanding, it remains a higher return on investment than the market's long-term average performance level and, therefore, most desirable.

The only problem for an individual investor is that the fund is closed to outsiders. It is a company fund for the higher echelons of its management. It is understandably noteworthy that very few voluntarily leave the fund.

I point this out to show that higher return rates are achievable. If at least one can do it, you can do it too. Medallion Fund uses algorithms for its trading strategies, meaning software programs run the show.

The **One Percent Per Week** trading program, as presented in the above-published series of articles, uses the TQQQ ETF as its trading vehicle.

If we compare the TQQQ strategy results to the average market performance, it could be considered astronomical, even unprecedented.

But in light of Medallion Fund, those performance levels are in the same range. We should also compare those results to what the general market, on average, can provide over the long term, which is about 10% compounded. Whether it be the Medallion Fund or the TQQQ strategy, they both shine bright.

The big difference is that the TQQQ trading strategy is freely available to anyone who wants it. You can use it and modify its program code in whichever way you want.

The strategy has no minimum deposit requirement, minimum invested period, or preset exit protocols. Moreover, TQQQ currently trades in the vicinity of \$80 a share; \$100k would buy you 1,250 shares in an ETF trading about 60 million shares a day.

From the start, your limit orders would have no impact on the daily price movements of TQQQ. The point is that TQQQ is not out of reach; it is an ETF that is easily accessible.

On the TQQQ returns, that last cited-article, [High Stock Portfolio Returns? Easy](#), had CAGRs going from about 50%⁺ up to 100%⁺ and at each level, adding more alpha due to one thing or another. Nothing extraordinary. It was as if you do this, then it should do that with simulations and equations to justify those outcomes.

How could someone use a trading strategy made public more than two years ago and perform at such levels? Surprisingly, the strategy remained generally unnoticed⁴. Hopefully, this will bring more attention to it.

⁴ I am not the author of that strategy. I only slightly modified its code.

THE TQQQ PROGRAM

As its name implied, the program's original version aimed to make 1% profit per week. Its profit target was set at 1%, which was often hit, treating TQQQ as an ordinary stock with relatively lower volatility.

However, such a low-profit target was not adapted to what the strategy was trading.

Moreover, because the original strategy wanted to buy with an initial discount, it aimed for stocks that opened while going down by at least -1% before the buy limit order could be executed – the other part of the reason for the strategy's name.

In doing so, the strategy would ignore most of the rising part of a price cycle and not profit that much from rising prices. It was initially programmed to do that, and it did.

Initially, that version of the program did not beat market averages. Understandably.

You could add a preset stop-loss procedure as downside protection to limit losses. But in doing so, the results would worsen.

BASIC EQUATIONS

Regardless of your age or where you're at, there are many solutions to answer your quest to exceed the growth rate of a savings account. All are choices you can and have to make.

But all will require time and capital, some with a bit of work and some a lot. Just like everybody else, you will not be able to extract blood from a rock.

Your quest for that higher growth rate will follow the future value formula, which states that your invested capital can grow at the average rate $\bar{g}$ over a specified time interval:

$$F(t) = F_0 \cdot (1 + \bar{g})^t \quad (1)$$

It does not matter what you intend to invest in; the equation will prevail. Supply your numbers and evaluate the outcome. Will it be enough, and would you be able to do it? Do you have the initial capital and determination it takes to get it started? Do you have enough time?

You are 45 years old and consider your retirement investment to have two phases, one before retiring and the other after. The following equation could represent your long-term objective:

$$F(t) = [F_0 \cdot (1 + \bar{g})^{20}] \cdot (1 + \bar{g} - 0.05)^{t-65} \quad (2)$$

where you have 20 years before retirement to grow your fund. From $t > 65$, you could withdraw 5% per year from your still accumulating fund.

It would all depend on the value of $\bar{g}$, the average growth rate of your investment portfolio. For example, if $\bar{g}$ is less than 0.05, your fund will shrink $(1 + 0.03 - 0.05) = (1 - 0.02) = 0.98$. After 20 years, your portfolio will have shrunk to 0.6676 of the value it had at 65, and your withdrawals would also have shrunk by -33.24%⁵.

No matter how you would want to cut it, it will all come back to equation (1) or (2) and their parameters:

- How much capital (F_0) do you have? Can you find more?
- How much time (t) do you have before retirement?
- What kind of growth rate ($\bar{g}$) do you think you can achieve over the period?

You are not limited to a single investment; you could do many. That is simple too. You add them all up as you go:

$$F(t) = \sum_1^N f(t_i) = \sum_1^N f_{0i} \cdot (1 + \bar{g}_i)^{t_i} \quad (3)$$

Many investments come with a preset $\bar{g}_i$, like your savings account, a coupon bond, or the market's expected long-term average return.

Buying SPY for the long term will not make you exceed SPY's long-term CAGR.

Its long-term average growth rate will remain near $\bar{g}_i \approx 10\%$, on the condition that you stay fully invested for the duration (that is 20 years or more). Still, your capital would only be expected to grow at SPY's long-term average growth rate, but no higher.

As long as we do not see some of the available higher return options, we consider what is offered as the best available.

We usually don't question beyond what is offered. "They" will tell you that you need a professional (like them) when you could do it all yourself with a much better outcome.

You see Medallion Fund with its high-performance level, and know you could do that too if you knew how. Until you realize you could do something different and obtain even better results.

How difficult could it be to buy and sell shares?

In my articles, [There Is Always A Better Retirement Fund - Part I](#), and [Part II](#), it was shown that there is always a better trading strategy. It is not because you found a better one that it is the only one.

⁵ A reference to equation (3) in [Your Investment And Retirement Plans](#).

There usually are many many more since even those better strategies have siblings that can do even better. In fact, there is an infinity of them. But all you need are just a few. You have limited capital, after all. Only after finding a better strategy can you consider replacing an older one or start looking for an even better one.

No matter your investment methods, you will have a starting and ending point in time. To determine your average growth rate, rearrange the future value formula (equation (1)) to extract $\bar{g}$:

$$\left[\frac{F(t)}{F_0} \right]^{\frac{1}{t}} - 1 = \bar{g} \quad (4)$$

The above equation applies to past or future data points.

Knowing we had a 10% return over the past 20 years after buying SPY, we could use that CAGR to make estimates of where your investment is going. For instance, adding 5 years to the future value equation (1) would give:

$$F(t) = F_0 \cdot (1 + \bar{g})^{t+5} \quad (5)$$

We would have a continuation of what was for an added 5 years.

When applied to an exponential function, adding 5 or 10 years could make an enormous difference depending on the value of t , F_0 , and $\bar{g}$.

We can make the above estimate because we already have a twenty-year history to back up that claim (equation (5)).

The estimate made becomes what might be considered a reasonable outcome. It does not assure that it will happen, but it might be the most reasonable estimate without other knowledge to prove or disprove it.

It is like with the S&P 500, which since 1957 has had a long-term $\approx 10\%$ average return. It is higher than the total market average, as should be expected. After all, it looks only at the top 500 out of the thousands of stocks in that investment universe.

Furthermore, you had 43 years of data before 2000 to predict where the future SPY might end up. The rate of return of SPY is going the same way as the S&P 500 index since it tracks its return daily.

Therefore, from 2000, the expected outcome would have been to have your SPY investment grow to $F(2024) = F_{2000} \cdot (1 + 0.10)^{24.8} = 10.63 \cdot F_{2000}$. The SPY portfolio would have increased 10.63-fold its original value. Starting with \$100k, your investment over those 24.8 years would have exceeded \$1 million.

It is a good enough approximation, especially when making such a prediction from 24.8 years ago. That SPY estimate was based on historical data, and the future expected return would hold as that past return has held over the prior years.

That S&P 500 index average return ($\approx 10\%$) is also the long-term average return achieved by the average money manager. It is setting kind of average limits. You have the best out there, and they have a hard time beating market averages.

What are your chances to do better? Why even consider that you could?

You can get market averages almost for free; you buy and hold SPY for the duration, and job done.

Therefore, your first move should be to buy an index fund; this way, you could get close to the market's long-term average return. Saying you could buy and lock that long-term rate of return on your investment on the condition you stayed invested.

But whatever, we should remain open-minded to different ways of doing things if they can prove themselves to be worthy of our attention.

We also need to decide if it would be appropriate for us to undertake such investment methods. It is all under our control. We are the ones making those decisions, and we are responsible for what we do.

We usually keep doing what we know best until we find something better.

Only after we find something better should we consider changing should we find it appropriate and desirable. We always have choices to make, and there is always a better investment strategy. As said previously, there is an infinity of them. We have to choose. We cannot spend 5 or 10 years searching for a better strategy; the opportunity costs are way too high.

The making of money in the stock market is not retroactive.

This TQQQ trading strategy is an oddity worth exploring.

TRADING TQQQ

There are common sense reasons why the **One Percent Per Week** trading strategy works so great when compared to other trading methods.

Gain Your Financial Freedom, will consider methods to raise $\bar{g}$ in equation (1) to propel you to much higher levels than a savings account and even much higher than market average.

That quest will still belong to you. My point might be: are you ready for the ride?

First, the easy solutions to outperform a savings account.

The easiest solution is to buy QQQ, SPY, DIA, similar alternatives ETFs, or any combination thereof.

Why pay an intermediary when all they would do, on average, is buy something equivalent to the constituents of these ETFs? Would their pick of 100 stocks be that different from another doing the same thing? Will you be going for the average closet indexer or average portfolio manager? How would you know which one you are going to get? They are as blind to the future as you are.

There are trillions of trillions of stock combinations using a selection of 100 stocks. When I did that kind of evaluation years ago, I had combination numbers in excess of 10^{400} . That number is monumental. Even if you made 10 trillion of those evaluations, each taking only a second, you would need 317,000 years to complete that job. and still have 10^{390} to do.

You are not in the sample portfolio evaluation business. You could not even review — in your lifetime — all those combinations to make what you might consider your best choice. But that would only analyze past data. Your future estimates would be based on what?

You want results, long-term results, and they better be above-average performance levels, or else you will fall short of your goal and should have at least considered the index fund solution.

Seek better investment solutions now so your future may generate more than average performance.

Instead of putting money in your savings account every week or every month or so, buy those ETF shares instead. Do this at every opportunity until you retire, as if contributing to your retirement plan.

You need a long-term view of the problem. Even doing this for 5 or 10 years might not be enough. Make the calculations to determine if it is sufficient for you or not. Reassure yourself that what you intend to do will help you reach your retirement goals.

The goal is that once you retire, you should have enough money to support yourself and your family for the next 30 to 40+ years. Add more as you go along; it will all pay off later. Keep buying these ETFs even after you retire.

Using SPY, your new "savings account" should get that $\approx 10\%$ long-term market average, which is already better than an ordinary savings account. I would add that it is even better than the average return on a government pension plan.

You would get an extra return boost by choosing QQQ over SPY or DIA. For example,

look up the article: [QQQ To The Rescue](#), where the compounded growth rate on QQQ is raised above $\approx 15\%$ over 24.5 years.

Selecting QQQ over SPY or a savings account is the same as choosing between a 5% return from a savings account

$$F(t) = \$100,000 \cdot (1 + 0.05)^{24.5} = \$330,474$$

or a 10% CAGR using SPY

$$F(t) = \$100,000 \cdot (1 + 0.10)^{24.5} = \$1,033,048$$

or a 15% CAGR using QQQ

$$F(t) = \$100,000 \cdot (1 + 0.15)^{24.5} = \$3,069,708$$

It also puts a value for that single decision of selecting QQQ over SPY at more than \$2,000,000 over the period. A choice you had to make. Which one would you have taken? Adding a zero to your initial capital would have generated 10 times more.

The same thing happens when you make other choices. Many of those decisions have opportunity costs we can express in dollars.

Would you have known in 2000 to select QQQ over SPY? I have to say yes. The difference was remarkably evident.

The top 100 of the top 500 best will overshadow the remaining 400 not selected, the top 500, and the overall general market containing thousands of stocks. It is like taking la crème de la crème, the rest is unimportant.

You should have been satisfied with the best of the group, knowing in advance that they would outperform SPY over the long term since SPY's average return would be dragged down by those other – less than the best – 400 stocks.

With no effort, we just raised the CAGR ante to a minimum of $\bar{g} \geq 0.15$.

A single decision and you outperformed the long-standing S&P 500 index market average. It still took 24.5 years to get there, and this is for whichever of the above three decisions you could have made.

By doing it yourself, you also retain complete control over your funds.

A serious advantage of doing this is that whenever you might need money, you can always sell shares and solve that problem without withdrawal penalties, excessive management exit fees, or even having to seek permission. You could even use your shares as loan collateral if you wanted to. It is very simple: it is always your money and your choices.

La raison d'être for TQQQ is QQQ.

QQQ has dedicated teams that evaluate, sort, rebalance, replace, and update its ETF on a continuous basis.

Why pay someone else to do the same kind of job? At least, you would get QQQ's long-term CAGR, which would surely be higher, on average, than a savings account and no worse than the average portfolio manager with their ($\approx 10\%$) average long-term returns.

Nonetheless, you could still be lucky and pick a portfolio manager that might exceed the 10% average return of the others. But there, your odds are about one in four since about 75% of them do not outperform long-term market averages. Which one would you have selected since you might only know after the fact which was which?

Buying SPY, on average, could double your portfolio every 7.29 years, while QQQ could do the same job in 4.97 years with no added effort.

Is QQQ more risky than SPY?

All of QQQ's stocks are part of SPY. QQQ is the top 100 stocks in SPY; therefore, it should be even less risky. QQQ's average is not dragged down by the other 400 stocks that are part of SPY. And the top 100 QQQ stocks are the most successful companies out there. If you look closely, you will find that those top 100 stocks in SPY are driving this ETF higher. Using QQQ, you went directly to the top 100.

*It took a lot from those companies to be in the NDX index,
and it will take a lot more to stay on it.*

Almost by definition, you are playing the "average" of the top 100 most successful stocks. It is not the same as playing a single stock. You are still playing 100 stocks but as a single ETF, as a single fund.

In November 2024, the TOPT ETF was released. It will be composed of the top 20 US stocks by market capitalization. Over the long term, it should exceed QQQ's return. You know this from the start. The top 20 of the top 100 of the top 500 of the thousands of listed stocks out there will outperform them all. That is not complicated. Since we have no data on TOPT, let's return to the problems at hand, but know that starting now, you could substitute QQQ for TOPT, increasing your future long-term CAGR.

Choosing QQQ over SPY is like buying a 5-alpha point advantage over the S&P 500 index and not paying a penny for that excess return.

We are often given that to gain alpha points; we need to earn them and bring more skills to the game. And here, we can buy some of those points with no effort at all, just by buying an ETF that comes with it.

By its very nature, QQQ is a little more volatile than SPY, meaning that QQQ could have larger drawdowns, but not by much. Is the added 5% CAGR points worth it?

Based on the calculation above, that QQQ decision is valued at more than \$2,000,000 and is a choice you have to make. Does it compensate for the slightly more bumpy ride? That depends on your views of what is acceptable to you.

If a stock goes down 40% or 45% before it fully recovers and then goes even higher, would it make that much of a difference that you would put a monetary value on it?

Selecting QQQ has better return prospects than your savings account or SPY. But again, it should not be enough. You should want more. You might be constrained by the limited time you have before retirement. You might have less than 40 years to build that retirement fund like a 25-year-old has.

So, adapt to what you can do within your time constraints, capital, and available and accessible investment alternatives.

Let's move faster and raise the ante once again.

We covered enough grounds to switch into high gear.

You could find, design, or modify a trading strategy that aims for returns higher than 20%. At that rate, on average, you could double your investment fund every 3.81 years following the sequence 2^n , which would give: 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, ...

That is the same sequence Mr. Buffett has had over the past 50-plus years. We cannot say it is impossible. He has done it. Many others have followed the same doubling sequence. So could you.

Buying QQQ would already get you closer to that objective. See the QQQ series of articles, especially [QQQ To The Rescue](#).

However, buying Berkshire Hathaway stock would have gotten you there. You could have had Mr. Buffett as your fund manager and not even pay a penny for the privilege and still get your average 20% CAGR.

Having invested \$100k in Berkshire Hathaway 24.5 years ago would have resulted in $F(t) = \$100,000 \cdot (1 + 0.20)^{24.5} = \$8,708,443$ per \$100,000 invested. And a lot more had you started earlier.

For instance, starting some 40 years ago, it would have given you: $F(t) = \$100,000 \cdot (1 + 0.20)^{40} = \$146,977,156$. Making a 25-year-old opened-eyed in wonderment and maybe with an expression like: "I could have done that too if I had the money". It

should be followed by: "Could I start doing that now?"

Could my parents have put me on the right track by starting it all for me while I was younger? What a gift that would have been! A lot of parents can do that now and prepare their child's future⁶.

Things are getting better and still based on single decisions you could have made. In 2000, Berkshire Hathaway and Mr. Buffett were already well known. He had been on the job for more than 25 years and has continued doing the same things he was doing before.

To raise your CAGR even higher, you will now have to bring guts, determination, and trading skills to the game.

THE ONE PERCENT PER WEEK PLAY

The **One Percent Per Week** trading strategy easily got to those higher levels.

It leapfrogged over a lot of trading systems in a single jump.

It will not matter so much where you get your CAGR or which strategy generated it as long as it becomes your CAGR.

And once you have a high CAGR performing strategy, exploit it until you find a better one.

While the **One Percent Per Week** strategy presents clear advantages, it's important to recognize that it's not just a single decision anymore but an entire trading ecosystem.

However, adopting this strategy as your preferred investment vehicle is still a single decision. Let's delve into the long-term implications and the potential benefits it could bring.

The strategy's foundation is the 3x-leveraged TQQQ ETF based on the QQQ ETF, which holds a portfolio of the top 100 highest-valued stocks on NASDAQ. Something that will not go away soon.

We could call that a solid foundation.

While part of QQQ, you will not see any of its stocks go bankrupt. Even if there were one, it would be one in one hundred. Furthermore, it would also be part of the

⁶ Look at the [The Big Open Project](#) for more information.

lowest-valued stocks on the list. In QQQ, the least valued stocks at the bottom of the list will usually be the ones to drop out or be removed, not those at the top⁷.

There will always be a top 100 highest-valued stocks, even if those on that list change occasionally. Also, you could always build such a list yourself should you not like QQQ's constituent stocks. But then, the high degree of correlation between TQQQ and QQQ might be lost, as well as its 3 times leveraging factor.

The stated objective of TQQQ is to deliver three times QQQ's daily return.

At the start of every trading day, TQQQ resets. This reset partly alleviates the otherwise expected long-term return degradation due to the higher volatility. Return degradation is faster for stocks with higher volatility.

For instance, the same number of $\pm 20\%$ moves will degrade more rapidly than the $\pm 10\%$ ones. Numbers come in as: $((1 + 0.10) \cdot (1 - 0.10))^{100} = 0.3660$ for the $\pm 10\%$ moves reducing one's portfolio to 36% of its original value. While the $\pm 20\%$ moves would give: $((1 + 0.20) \cdot (1 - 0.20))^{100} = 0.0168$, dropping the portfolio value by -98.32%.

The problem is deep-rooted and needs to be addressed.

Your trading procedures have to compensate for this phenomenon over the entire trading interval, knowingly or otherwise. It is pernicious since those $\pm\%$ return variations could be anywhere within the returns series: $\prod_i^N (1 \pm r_i)$. They do not have to be side by side, as given above, but spread out anywhere within the time series and in any order. It would have the same impact. And that is troubling.

The more trades increase, the more you will have those percent variations on both sides of the ledger ($\pm\%$). Also, the smaller the average price variation, the more will happen. There are many more short-term 5% moves in the market than 10% or 15% moves.

LET'S GET BACK TO TQQQ

We have had 14.4 years of TQQQ data to get some of its long-term portfolio metrics and some 25 years of the same on QQQ. TQQQ tries every day to triple QQQ's return and usually gets pretty close to it.

You had ten-plus years of QQQ data (started in 1999) before TQQQ's release in February 2010. We had more than 500 weeks of weekly QQQ data to extract useful information before the release of TQQQ.

⁷ Refer to the first chart in [Take the Money and Keep it – II](#) which show weights of less than 1% for the least valued stocks in the group.

You already knew QQQ's average long-term metrics before getting to TQQQ.

From the start, we knew TQQQ would have **triple** QQQ's daily return and volatility.

So, no surprises there. The average weekly volatility for QQQ gets close to 5%, so the expectation for TQQQ's volatility would be about 15% per week. Also, QQQ's return would average about a 15% CAGR⁸, making TQQQ's expected average return aiming for a 45% per year.⁹

By selecting TQQQ as a trading vehicle, we have upped the CAGR to an expected long-term average of $\approx 45\%$. Nonetheless, as expressed previously, we need to make provisions for the expected long-term return decay.

We should expect that even with all the price gyrations, and as the number of weeks grows, the 15% average volatility, or thereabout, will be maintained going forward. Of note, volatility increases with time, suggesting that the ride might be bumpier in the future.

LEVERAGING TQQQ? YES

Table #1 below was constructed using my version of the TQQQ trading strategy. It has a few minor modifications to the original program, found in the Wealth-Lab 8 examples section and on its website.

Table #1: The One Percent Per Week Trading Strategy With Leverage

Leverage Factor	Outcome (\$)	CAGR (%)	% Per Trade	Win %	Loss %	# Wins	# Losses
1.0x	72,792,124	56.94	1.02	4.54	-2.71	393	370
1.1x	114,511,095	61.87	1.02	4.54	-2.71	393	370
1.2x	175,357,154	66.65	1.02	4.54	-2.71	393	370
1.3x	261,513,497	71.27	1.02	4.54	-2.71	393	370
1.4x	379,646,284	75.69	1.02	4.54	-2.71	393	370
1.5x	536,273,572	79.88	1.02	4.54	-2.71	393	370
1.6x	736,700,511	83.83	1.02	4.54	-2.71	393	370
1.7x	983,644,459	87.50	1.02	4.54	-2.71	393	370
1.8x	1,275,691,916	90.86	1.02	4.54	-2.71	393	370
1.9x	1,605,811,852	93.89	1.02	4.54	-2.71	393	370

The leverage factors for these simulations were the only change made before running the program. The outcome is the result of applying each factor from 1.0x for no leverage to 1.9x. Using other leveraging values would give intermediate results,

⁸ See [QQQ To The Rescue](#).

⁹ TQQQ has had 7 splits since inception (6 two-to-one, and 1 three-to-one). Without those splits and based on TQQQ's current price of $\approx \$80$, it would be around \$15,360.

such as having a 1.25x leverage factor, which would give something near halfway between the 1.2x and 1.3x levels, as should be expected.

The last 5 columns in the table remained the same for all scenarios, illustrating that only the position size was changing, not the average return percentages.

Should you modify the program as I did, which I most highly recommend¹⁰, you should get the exact same results over the same time interval I used. That was over all the TQQQ available data since its inception in 2010 up to the simulation date: September 28th, 2024.

We should remember that the **One Percent Per Week** program aims to generate 1% per week. Its stated objective is to produce: $(1 + 0.01)^{52} = 1.6777$ or 67.77% per year.

I used the **One Percent Per Week** trading strategy to make all the simulations using the latest data with \$100k as starting capital. I recorded the results at each leverage setting, going from 1.0x to 1.9x.

The leveraging factor is set outside the trading script before we can run the strategy.

There is no hype in those results.

The results of the simulations were the outcome of executing the same program procedures as coded. There was no curve fitting, no looking at forward data, no parameter optimization, no survivorship biases, no fudging of numbers, and no cheating of any sort.

Taking a trade every Monday at the open cannot be considered any of those things. Furthermore, all the trading procedures used are the application of common sense.

Nonetheless, it could be considered a bet to the upside. To make a profit, the price will have to rise over the next 2 to 5 trading days, and this before the closing price on the coming Friday.

Some observations on the above table:

- The only change to those simulations was the leveraging factor
- As the leverage increases, the outcome increases as should be expected
- The strategy's CAGR also increases in step with the outcome
- The leveraging fees also increase with the increasing factors
- We can see a slight slowdown in the CAGR rate, but it remains impressive
- The metrics on the executed trades remained the same for all leveraging levels
- the average percent win per winning trade remains at 4.54%

¹⁰ I decided to include my version copy of the program in the related files section.

- · the average percent loss per losing trade stays at -2.71%
- · the number of winning trades and losing trades for all levels are the same
- · the average percent gain per week remained the same at 1.02% per trade

In Table #1, these outcomes were after having paid the leveraging fees, which got more significant as you raised the leveraging factor but represented only a small fraction of the outcome. It is sufficiently low that we should consider the leveraging fees as the cost of doing business, especially in light of those higher CAGRs.

For instance, at the 1.5x leverage factor, the paid-out leveraging fees were about 4.3% of the total profit generated when the strategy's CAGR came in at 79.88% after having paid those fees.

We could say we paid for or purchased those higher returns. Look at any business; they all would gladly pay a total of 4.3% in fees from their potential profits to get a 79.88% return on their investment. Perhaps so would you.

For instance, stakeholders in Medallion Fund pay 4% in fees per year and 44% of profits and be assured they do not complain about high fees.¹¹

Another important point in Table #1 is that the overall CAGR increases as the leveraging factor rises. It goes from 56.94% for no leverage to 93.89% at the 1.9x leverage net of fees.

Such growth rates surpass the net 39% CAGR of Medallion Fund. The TQQQ strategy's results are net of leveraging fees, which makes the outcome even more valuable. You do not have to pay out 40% of the profits to anyone. It is all yours.

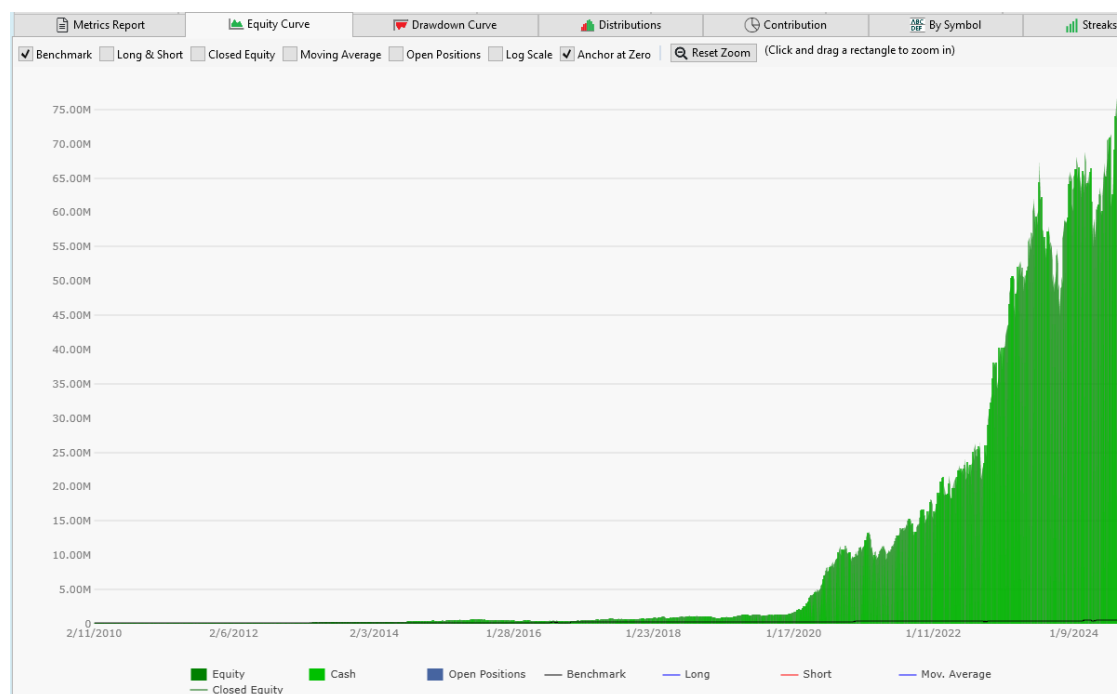
Depending on your determination and the speed at which you would like to operate, it gives you a roadmap as to how aggressive you might like to be. Nonetheless, it remains in the realm of your choices, and you will be the one making those decisions. If you do not like leveraging, take the 1.0x level, which uses no leverage.

All the simulations were over the same time interval. To accelerate your outcome, you could raise your leverage factor to accommodate the time horizon you have. Table #1 needed 14.4 years to do its job. It is not an instantaneous outcome. You will have to work for it. That is only a little work, even if done by hand (a few minutes a week). Or, let your computer program do the job.

Let me display the result of the initial simulation with no leverage (Figure #1 below), which will corroborate the first line in Table #1, showing that the simulation was an actual outcome of that program.

¹¹ The mentioned 4.3% in leveraging fees is not per year but the total over the entire trading interval.

Figure #1: TQQQ Strategy - Equity Curve - No Leverage



[\(Click here to enlarge\)](#)

Anyone could redo all the simulations in Table #1. You can find the modifications I made to the program in the first two of the **One Percent Per Week** article series¹².

There are no mysteries in the presented charts. If you do not get the same results as displayed in Figures #1 and #2, the reason is that you have yet to make the simple recommended modifications given in the two above-cited articles. So, back to your homework.

The first program modification changed the profit target from 0.01 to 0.07. It required changing a single character.

The second modification said not to wait for the -1% discount to take a trade on Monday but instead buy a penny above the open, requesting a show of strength before taking a position. That was easy to do too: changing the entry condition from $(< -1.0\%)$ to (> 0.0) .

I saw a diamond in the rough when I first looked at that published strategy. A few cuts here and there were sufficient to show it was a real gem of value.

My modifications to the program changed its very nature and trading behavior.

The program no longer waited for the price to fall before entering a trade. I raised

¹² I have also provided a link to my version of the program in the related files section.

the profit target to 7%. If the price was already doing fine, meaning on its way to the 7% profit target, it was raised by 1% more. A move that paid off, as described in the series of articles.

The faint, almost horizontal line at the bottom of Figure #1 is SPY, which was taken as the benchmark. Based on Figure #2, with no leverage, TQQQ generated \$72,792,124 over those 14.4 years and a 56.94% CAGR, while SPY did \$433,824 giving it a 12.16% CAGR. Both started with \$100k.

Figure #2: TQQQ Strategy - Portfolio Metrics - No Leverage

Analysis Series	Rolling Profit
Metrics Report	Equity Curve
Select ScoreCard: Basic ScoreCard	
Strategy	Benchmark (SPY)
Summary	
Starting Capital	100,000.00
Profit	72,792,124.43
Profit %	72,792.12%
Profit Per Bar	15.25
APR	56.94%
Std Dev of Annual Ret...	164.30%
Exposure	51.44%
Maximum Exposure	99.91%
EAR	110.68%
Alpha (α)	39.78
Beta (β)	1.47
Sharpe Ratio	1.27
Sortino Ratio	2.18
WL Score	50.39
Slope of Equity Curve	13,119.52
Positions	
Position Count	763
Avg Profit	95,402.52
Avg Profit %	1.02%
Profit Factor	1.62
Payoff Ratio	1.67
Avg Bars Held	3.35
Avg Trades Per Month	8.67
Avg Bars Held as % of...	0.09
Largest Bars Held as %...	0.14
NSF Position Count	0
NSF Ratio	0.00
Drawdown	
Max Drawdown	-22,726,223.93
Max Drawdown Date	10/26/2023
Max Drawdown %	-54.47%
Max Drawdown % Date	7/6/2010
Recovery Factor	3.20
Profitable Positions	
Count	393
% Profitable	51.51%
Avg Profit	483,537.61
Avg Profit %	4.54%
Average Bars Held	3.03
Unprofitable Positions	
Count	370
% Unprofitable	48.49%
Avg Loss	-316,859.88
Avg Loss %	-2.71%
Avg Bars Held	3.70

([Click here to enlarge](#))

Starting with \$1,000,000 would have generated 10 times more for both scenarios: SPY and TQQQ.

Getting that \$1,000,000 could increase your performance 1000% before you even started playing the game.

It does put a value on your efforts to raise more as initial capital. Note that capital could be added to the strategy whenever you choose. Your long-term outlook would

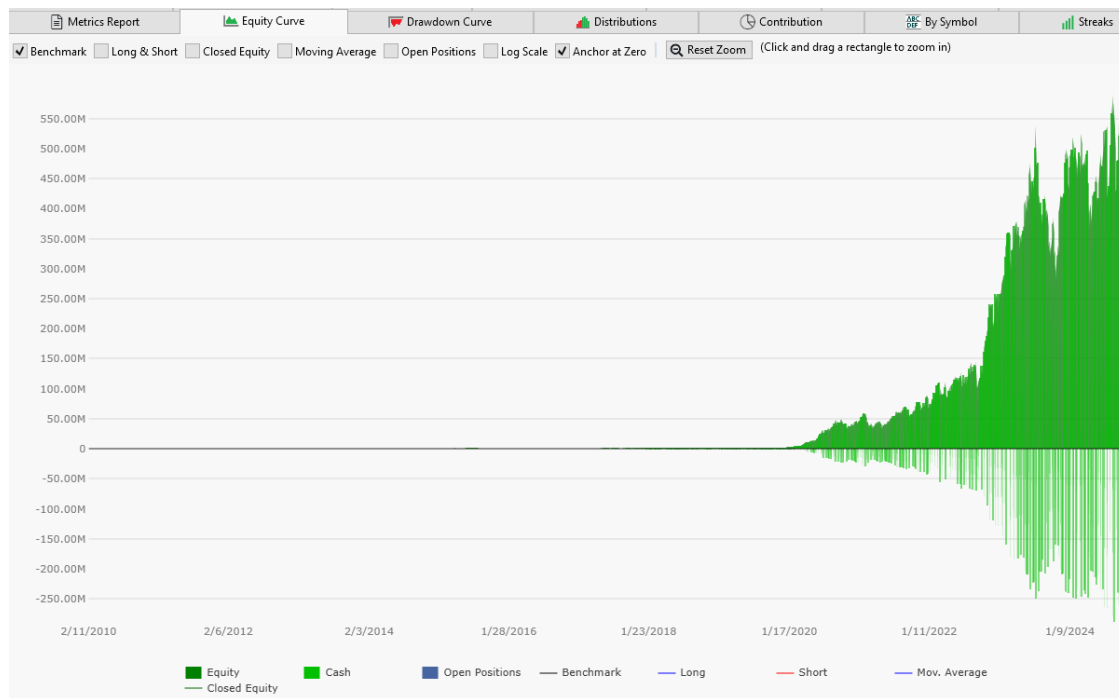
only improve.

The portfolio metrics from Figure #2 list the averages to be used later. These averages will serve as long-term averages that had 14.4 years or 763 trades to settle down.

The TQQQ strategy is not for everyone. But still, do consider what is possible. Knowing it is easily achievable, you could also design other trading strategies to reach those high CAGR levels. You intend to play a game of money where you need the funds to put on the table. That is also simple: no funds, no game, no profits. The only bright spot would be no loss, either.

For what is coming next, I will use the 14.4-year metrics in the table above, which were extracted from Figure #2.

Figure #3: TQQQ Strategy - Equity Curve - 1.5x Leverage



[\(Click here to enlarge\)](#)

Figures #3 and #4 also corroborate the data in Table #1. This time for the 1.5x leverage setting. We have the equity curve, SPY, and a show of the leverage in Figure #3 (light green bars below the zero line) and the corresponding portfolio metrics in Figure #4.

Portfolio Metrics

Portfolio Metric	Description
56.94%	average percent portfolio return
51.44%	average exposure rate
51.51%	win rate average
763	total number of trades
393	number of winning trades
370	number of losing trades
1.02%	average percent profit per trade
4.54%	average profit percent per winning trade
-2.71%	average loss percent per losing trade

Figure #4: TQQQ Strategy - Portfolio Metrics - 1.5x Leverage

Analysis Series

Rolling Profit

Metrics Report

Equity Curve

Select ScoreCard: Basic ScoreCard

	Strategy	Benchmark (SPY)
Summary		
Starting Capital	100,000.00	100,000.00
Profit	536,273,572.78	433,824.32
Profit %	536,273.57%	433.82%
Profit Per Bar	15.25	5.90
APR	79.89%	12.13%
Std Dev of Annual Ret...	455.79%	13.62%
Exposure	77.75%	99.96%
Maximum Exposure	174.59%	99.99%
EAR	102.75%	12.14%
Alpha (α)	54.57	-
Beta (β)	2.22	-
Sharpe Ratio	1.21	0.77
Sortino Ratio	2.21	1.13
WL Score	25.36	8.00
Slope of Equity Curve	87,742.45	101.10

Select ScoreCard: Basic ScoreCard

	Strategy	Benchmark (SPY)
Positions		
Position Count	763	1
Avg Profit	732,879.78	433,824.32
Avg Profit %	1.02%	434.13%
Profit Factor	1.48	-
Payoff Ratio	1.67	-
Avg Bars Held	3.35	3,681.00
Avg Trades Per Month	8.67	0.01
Avg Bars Held as % of...	0.09	99.97
Largest Bars Held as %...	0.14	99.97
NSF Position Count	0	0
NSF Ratio	0.00	0.00
Drawdown		
Max Drawdown	-255,435,828.38	-113,154.10
Max Drawdown Date	10/26/2023	10/12/2022
Max Drawdown %	-75.32%	-34.10%
Max Drawdown % Date	8/22/2011	3/23/2020
Recovery Factor	2.10	3.83
Profitable Positions		
Count	393	1
% Profitable	51.51%	100.00%
Avg Profit	4,360,532.04	433,824.32
Avg Profit %	4.54%	434.13%
Average Bars Held	3.03	3,681.00
Unprofitable Positions		
Count	370	0
% Unprofitable	48.49%	0.00%
Avg Loss	-3,120,275.18	-
Avg Loss %	-2.71%	-
Avg Bars Held	3.70	-

(Click here to enlarge)

THE PORTFOLIO'S EQUATION

Defining a stock trading program with explicit and concrete equations is unusual.

But here, we already have an equation that represents this trading strategy. It also appeared in my prior articles for years. So, again, nothing strange or unheard of.

Since the last walk-forward was ten weeks ago, the portfolio metrics with no leverage applied got updated to their current values (see the new numbers in Figure #2 above):

$$0.50 \cdot \$100,000 \cdot (1 + 0.0454)^{393} \cdot (1 - 0.0271)^{370} = \$72,815,878 \quad (6)$$

The simulation, with a leverage of 1.0x (meaning no leverage), came in at \$72,792,143. The rounding of return rates can account for the difference in outcome. For example, with the 4-decimal rounding for the range 0.0453501 to 0.0454499 in the percent win per winning rate, we could have an outcome difference of \$2,732,071. Thereby stating that the calculated outcome approximates the results and is subject to the rounding of values.

If we apply a 1.5x leverage factor, the answer is not as simple as:

$$1.50 \cdot 0.50 \cdot \$100,000 \cdot (1 + 0.0454)^{393} \cdot (1 - 0.0271)^{370} = \$109,223,817 \quad (7)$$

Since that is not what we got. See the profit metric in Figure #4.

The outcome came in at \$536,273,572.

The reason for the difference in outcomes is very simple.

Evidently, it was not only the first trade that was compounding. It was every bet made f_{i-1} , from $i = 1$ to N .

The leverage should have been applied to every bet – as given in a prior article – we should have used the following equation:

$$F(t) = \sum_{i=1}^N f_{i-1} \cdot (1 \pm r_i) \quad (8)$$

where f_{i-1} is the bet made using the portfolio value from the previous Friday's close. We have for $i = 1$, $f_{i-1} = f_0$, the initial bet made. The bet size will grow at the one-week delayed exponential rate, in step with the portfolio's rising value. It also means that the result will swing around depending on the return of each bet $\pm r_i$.

The problem we have is that we do not know the future value of those N bets. We know the first one in the series, but not necessarily the $N - 1$ that follows.

We could make estimates, but that would require making $N - 1$ of those f_{i-1} and for $i = 2$ to N before getting an expected portfolio value $E[F(t)]$.

Regardless, that would be a totally futile exercise. Who knows the value of their portfolio one, two, five, or ten years from now? You would soon figure out that – wishing estimates for what could be, unburdened by what has been – have little value, to paraphrase someone.

Even Artificial Intelligence (AI), with all it can do, would not be apt for the task. Predicting random-like price variations is not AI's forte, nor is it mine. No AI machine will be able to outsmart randomness. That is, to outsmart the flip of a fair coin.

If you want to use AI in your strategy design, make it part of the stock and trade selection processes, where it might help. Design an expert system that tries to anticipate the future of companies and their interaction within their ecosystems.

Then, let your AI define and isolate trends. Your objective would be to raise the win rate above 0.60 and have the average percent win per winning trade greater than the average percent loss per losing trade. The same objectives as in the **One Percent Per Week** strategy but at a higher level and with the potential of generating an even higher long-term CAGR.

With the **One Percent Per Week** strategy, a big structural difference comes into play, and it has consequences.

We have an equation that describes how we got there, meaning explaining $F(t)$, the outcome of our trading program. And this for 14.4 years of trading over what should be considered randomly distributed price variations.

Even with the high degree of randomness in play, we have an equation for the outcome of the TQQQ trading strategy.

With equations (1) and (5), we could make projections based on the long-term averages characterizing the inner workings of the strategy. These were set in equation (6).

Table #1 shows that essential portfolio metrics stayed the same following a change in the leveraging factor. This included the average profit per trade, the number of winning and losing trades, and their respective average percent profit or loss.

None of the trades were affected by the $(1 \pm r_i)$ section of equation (8) as given in Table #1. And, again, justifying the use of equation (6) as a reasonable representation of the portfolio's outcome.

The long-term percentage win per winning trades has had 763 weeks to settle down to 0.0454, as in equation (6) above. For example, should the next trade generate a 5% return and be added to the long-term average of 0.0454; the average would increase to: $0.0454 + \frac{0.05 - 0.0454}{764} = 0.0454 + 0.0000060 = 0.0454060$.

The average long-term percent win per winning trade would oscillate near this average since trades are only slightly biased to the upside (51.51%). Moving the average away from its ongoing long-term mean will get harder and harder. Furthermore, the long-term average volatility is limited to what can happen during a single week since all trades are closed on Friday's close at the latest.

Nonetheless, due to the upside bias, the long-term percent win per winning trade would slowly, very slowly, increase as more and more trades are added. It would take quite a few of the 0.0000060 increases to be noticeable. But their impact will be visible over the long term.

Not only does the Law of large numbers come into play here, but also the tendency for these averages to mean revert over the long term. We will have small variations, up and down, as we add more and more weeks to the strategy, like the one-week move of 0.0000060 above, which will have little impact. Furthermore, the overall effect of that particular trade will continue to decrease as we increase the number of trades. Again, following the Law of large numbers.

All the portfolio metrics we will use will have had 763 weeks to settle down to their long-term averages. As such, these averages could also represent their future expectation and respective values.

We will use the knowledge acquired from the strategy's behavior to set forward guidelines as if saying that the strategy could continue to do what it has been doing for the past 14.4 years of weekly trading.

A GAME CHANGER

Equation (8) is a game changer. We should update Equation (6) for the 1.5x leveraged scenario to see its outcome:

$$0.50 \cdot \$100,000 \cdot (1 + 0.00262)^{763} \cdot (1 + 0.0454)^{393} \cdot (1 - 0.0271)^{370} = \$536,131,700 \quad (9)$$

As expected, it does corroborate the simulation result for the 1.5x leveraging factor (see Table #1 and Figure #4).

The position size is growing at a 0.262% rate per trade. It does not seem like much, like 26 cents on \$100 dollars. But still, it managed to turn \$100k into \$536,273,572 in 14.4 years, net of leveraging fees.

That should be considered a feat, with an overall CAGR of 79.88% over the period after paying all the leveraging fees. Thereby even eclipsing Medallion Fund by a wide margin.

It is not you might, but you can do better.¹³

The long-term 10% average market return would give 0.0019 or 0.19% per week. Yet, we have the bet size increasing at 1.02% per week based on Table #1, and confirmed by the average profit percent metric of Figure #4. It might not be that big of a rate difference, but it is a big enough difference in the outcome of the strategy.

To further make the point. Raising the leverage to 1.7x should generate:

$$0.50 \cdot \$100,000 \cdot (1 + 0.00342)^{763} \cdot (1 + 0.0454)^{393} \cdot (1 - 0.0271)^{370} = \$985,295,913 \quad (10)$$

The strategy at this 1.7x leverage, according to Table #1, came in at \$983,644,459, making the equation a reasonable estimator of the outcome. And again, validating the structure of equations (8) and (9).

We got our numbers from the performed simulations, and they explain and justify using the above portfolio equations.

A LOOK AT THE FUTURE

Market simulation offers a powerful tool to assess the potential profitability of a trading strategy based on past data.

But there is no money to be made there. All we can get back is information.

Only what was, unburdened by what is or will be. Nonetheless, that information can be valuable.

A working program — that does not crash, even with bugs —, is not the same as having a program that will work in the future and deliver results.

Any trading script needs to exploit stuff that happens in its market environment. We set the trading rules and conditional thresholds for making trading decisions.

It is like designing your own game within a game, setting your own rules of engagement for your participation. You intend to play your game by your rules within that market trading game while following all its regulations and constraints, theirs and yours.

Your set of trading rules will make a difference.

¹³ See **There Is Always A Better Retirement Fund: [Part I](#), and [Part II](#).**

They do not need to be complicated. But if you prefer complicated, go for it. The stock market is a very unforgiving and complicated ecosystem where anything can happen.

It is your job to navigate this future sea of variance with all its tumultuous, random-like waves of unknown amplitude.

Your trading rules are like a maze you put out for prices to follow or else. You are out with a profit if the price hits your profit target. As simple as that. Otherwise, you will wait to exit your position on the coming Friday's close with a lower profit than your target or a loss. But you will know which will happen only once it happens.

**The market does not know what you will do
and keeps no memory of what you have done.**

**It is not in the past that we need to make money.
It is in the future.**

And there, we are almost blind as to what will or might happen. No matter the statistics you put on the table, there is no assurance they will continue to hold in the future.

The world is constantly changing. For whatever you wish for the future, it will be a bet you make that it might come true, and most often, with no certainty. "You are going to win if you do not lose" seems to be the mantra, especially in a game where your winning odds are 51.51% or thereabout. It is as if morphing into a game played with a slightly and positively biased coin.

We have statistics to help us determine what is probable since historical data shows that weekly percent price variations tend to long-term averages. We can also prepare for events we know are most probably coming our way.

It is like Mr. Buffett's bet on America. He decided early that the US would prosper and opted to participate market-wise, believing that the best companies would continue to grow. Would you make the same bet today?

Berkshire Hathaway's top stock positions are similar to the top stocks part of QQQ. Both could be samples of what we can put in our portfolios.

The TQQQ ETF is like putting Berkshire Hathaway or QQQ on steroids. You are going for a 3x-leveraged scenario on the best, where you do not have to pay leveraging fees unless you opt to leverage even higher by leveraging TQQQ itself.

The coming estimates (see Tables #2 and #4 below) are based on the long-term metrics of the TQQQ strategy.

The simulation had 763 trades to reach those metrics over a 14.4-year trading interval. That number of trades is more than sufficient to say those metrics were statistically significant and representative of their long-term averages.

To put this in perspective, if it were one trade per month instead of one trade per week, it would take 63½ years of data to get that high a number of trades, and 763 years if we were dealing with one trade a year.

The TQQQ trading strategy operates on a weekly basis. It lives in the highest variance space of annualized returns¹⁴.

Equation (8) is compounding on a weekly basis. Still, its weekly returns are quasi-randomly distributed. The $\pm r_i$ might be small for most of those trades, but the sum of their outcomes, profits, and losses accumulate over time due to the trading procedures used.

On the same assumptions, we can make long-term future estimates on the S&P 500. We can also make forward estimates on the TQQQ ETF. We have more than a sufficient number of trades to make the statistics we have accumulated significantly representative of the whole.

Furthermore, TQQQ is based on QQQ, which is based on the top 100 stocks in the S&P 500 index. Therefore, if we can make forward estimates on the S&P 500, we can do the same on either QQQ, SPY, DIA, or TQQQ. Nonetheless, those returns $\pm r_i$ will make the results fluctuate with time but still tend to their long-term averages.

The difference between making estimates on the S&P 500 or TQQQ is that the latter is, at least, 3 times more volatile than the S&P 500 index. That should be expected without making any other comment; TQQQ, by definition, was constructed that way.

Since TQQQ is designed to reset every day and generate 3 times QQQ's daily return, we can count on TQQQ to generate 3 times more than the S&P 500. And since QQQ has a long-term CAGR higher than SPY, we should expect that TQQQ's return will be higher daily than 3 times that of SPY. By leveraging QQQ, you also leverage its free 5-alpha point generation 3 times, as expressed earlier.

Also of interest is the notion of simulations. Some will say that simulations are worthless. Usually, we hear this from people who are not doing any, meaning discretionary short-term traders. But, here is the thing.

A discretionary trading method, if consistent, can be programmed. We could implement any trading procedures a discretionary trader can have in software. We can send people to the moon and can certainly program any of your trading rules,

¹⁴ Refer to [The One Percent a Week Stock Trading Program - Part VII](#) for more explanations.

whatever your reasoning might be.

My point is that if your discretionary trading rules have any value, they should stand the rigors of simulations over past market data. Otherwise, what is the evidence or proof that your "discretionary" methods have any future value?

Without a simulation phase, you would not send a spacecraft to the moon or fly a plane with people on it. If your "discretionary" trading methods cannot survive a 10 to 20-year simulation, how could you have any confidence in your trading methods to last that long? If your methods cannot provide a decent return on past market data, they will have no supporting evidence or any reason to succeed in the future.

The trading landscape universe is literally filled with failed discretionary wannabe short-term traders. But then again, if it is your choice, go for it. However, do not tell me you have proof by showing only one good example. I will need more than that. Your gambling record might not be enough. The exceptions I would make – on the discretionary thing – would be for the long-term investors where the odds of winning that game are much better and positive.

FORWARD ESTIMATES

Using the above equations and assumptions, I made simulation estimates for 5 and 10 years forward. The assumption is that the long-term averages will be maintained or stay close to their respective long-term averages.

We are not looking for an exact number; we are interested in a reasonable approximation of what might be. We could also make hundreds of simulations for ranges of values, but we do not need that to understand the workings of this strategy.

We will use the long-term portfolio metrics we acquired over time to make estimate approximations that might hold true going forward. It's like having the strategy telling us what it will do in the future since it did the same things so often in the past.

We could not make those estimates if we only had a few trades. The sampling errors would be too high, and the deviation from the mean even more untrustworthy. We need large samples to make relevant estimates, a condition easily met with the 763 weekly trades of this strategy.

The most reasonable bet and assumption is for a continuation of those long-term averages, which should not venture far from their 763-trade averages. It is the same assumptions we make over the long-term appreciation rate of SPY, which holds near the 10% market average, or for Berkshire Hathaway with its 20% average or QQQ and its 15% average. They are more likely to continue doing what they have done over the years.

On average, the market has trended upward for the last 200⁺ years. Your strategy could also trend up if sufficiently general and under the same principles primarily because TQQQ is based on QQQ, which holds the top 100 stocks of the S&P 500.

In Figures #2 and #4, SPY's average return over the 14.4-year trading interval was at 12.13%. But nothing is stopping it from reverting to its long-term mean. There is also no reason to assure anyone that it will revert to its long-term mean either, only that it is most likely to do so. It could continue doing what it has done over the past 14.4 years.

Regardless, we want higher returns than SPY's average long-term return.

Here are the assumptions made for these estimates based on the TQQQ strategy's 14.4 years of weekly data and its trading rules.

It will be assumed that:

- There will be, in the future, more Monday opening prices
We do not know those future prices; none of them
But we know those trades will be closed by Friday's close
- Friday's close will be there too, just as Monday's open
We do not know those prices either; again, none of them
- Before the week is over, a profit target might be hit
The profit targets will not change value
If executed, your 7% limit sell order will give you a 7% profit
The 8% profit target would do the same

All the above points will be there in the future, just as they were in the past.

The portfolio's equation trade metrics might not change much.

We have not changed the past and will not change the future.

In portfolio equation (6), the section: $[(1 + 0.0454)^{393} \cdot (1 - 0.0271)^{370}]$ as illustrated in Table #1 will not change even if you leverage the portfolio. Based on the TQQQ trading rules, that should have been expected.

The simulations using leverage showed that that equation section stood its ground. All trades for all the leveraged scenarios had the same metrics (% average profit per trade, % profit per winning trades, % loss per losing trades, # of wins, and # of losses).

We can rewrite equation (6) and factor that section out:

$$F(t) = \bar{e} \cdot b_0 \cdot (1 + \gamma)^N \cdot [(1 + 0.0454)^{393} \cdot (1 - 0.0271)^{370}] \quad (11)$$

where γ is the growth rate of the position size, b_0 is the initial bet (capital) put to work, and N is the total number of trades: $N = 393 + 370 = 763$. And where $\bar{e}$ is the strategy's average exposure rate.

We need to figure out γ for each level of leverage. Here, the simulations on the past data will provide those γ numbers for the past 14.4 years for each leverage level we might consider. It should be why we do simulations in the first place: to get those long-term averages.

We need to figure out γ for each level of leverage. Here, the simulations on the past data will provide those γ numbers for the past 14.4 years for each leverage level we might consider. It should be why we do simulations in the first place: to get those long-term averages.

It is like having the trading strategy tell us how it behaves, whatever market conditions are thrown at it. On average, it will do this as it has done in the past.

We already know the outcome of those leveraged scenarios based on Table #1. We have done all those simulations. Extracting γ is easy. It's the only remaining unknown, and the leveraged simulations prove that those γ rates exist since they are the solution to the portfolio equation.

SIMULATING THE FUTURE

We will use the same TQQQ trading strategy's long-term metrics and simple trading rules using the same initial capital: \$100k and the same initial time interval of 14.4 years as the starting point.

The idea is to build on what is there and use the acquired knowledge from those years, especially the long-term averages, which tend to mean-reverse and stay close to their long-standing average values.

We do these simulations to analyze the trading procedures we intend to use and see how they would have behaved on actual market data. Thereby gaining more insight into these software routines' inner workings and intricacies.

In real life, we would have to start from scratch, meaning from week one. The program would work and deliver according to its trading rules and thereby rebuild, week by week, a new outcome for the strategy.

But that is all in the future. The only way to verify any of that would be to live it. It is making any future estimate pure speculation.

Continuing what was, especially with the simple trading rules used in this trading strategy, can reveal much that could be applied, knowing where it might all go.

The outcome column in Table #2 is the result of the simulations performed to project the strategy's performance over an added 5 and 10-year period using the given parameters in Table #1.

Based on the strategy's average long-term metrics, I made estimates for an added 5 and 10 years. The lines with zeros in the minus trades column (–Trades) have the same outcome as in Table #1. We only need a general idea of the expected outcomes. So I limited the leveraging factors to 1.0x, 1.2x, and 1.5x.

Table #2. Five And Ten-year Projections With And Without Removed Trades

Leverage	Outcome (\$)	CAGR (%)	–Trades	Est. (T)	CAGR Est,	Est. T+5 years	Est. CAGR	Est. T+10 years	Est. CAGR
1.00	72,792,124	56.94	0	72,815,879	58.04	792,454,150	58.84	8,624,267,020	59.32
1.00	72,792,124	56.94	–10 Wins	46,709,061	53.24	508,334,033	55.25	5,532,191,905	56.45
1.00	72,792,124	56.94	–20 Wins	29,962,371	48.58	326,080,050	51.74	3,548,724,454	53.63
1.20	175,016,437	67.96	0	175,016,437	67.96	2,568,066,882	68.77	37,681,989,273	69.25
1.20	175,016,437	67.96	–10 Wins	110,984,509	62.73	1,628,507,847	64.85	23,895,567,385	66.12
1.20	175,016,437	67.96	–20 Wins	70,379,453	57.66	1,032,698,107	61.03	15,153,078,477	63.05
1.50	536,131,700	81.54	0	536,131,700	81.54	11,520,572,504	82.34	247,557,812,318	82.82
1.50	536,131,700	81.54	–10 Wins	335,029,380	75.71	7,199,220,381	77.98	154,699,191,150	79.33
1.50	536,131,700	81.54	–20 Wins	209,360,285	70.06	4,498,801,954	73.72	96,671,720,914	75.91

([Click here to enlarge](#))

A lot is going on in Table #2. We have an equation making market predictions 5 and 10 years out when we can hardly venture one for tomorrow.

All leveraging fees, if any, as in Table #1, were included in all the simulations except for the added years. Even though, theoretically, a significant part of it is included in the equation used.

For the added 5 and 10 years, the impact of leveraging fees was ignored even if most were technically included. I did not search for methods to extract those fees to evaluate how much were included, but I suspect most of them. They were already part of the equations used to make those projections.

Table #2 also considers the impact of removing 10 or 20 winning trades. We can see the outcome being significantly reduced even by removing only 10 or 20 winning trades out of the 763. We can see the impact and long-term effect of compounding on each position, no matter where they are in the return series.

The long-term impact on the outcome is understandable since the removed winning trades will not compound over the period. The effect on the estimates gets more pronounced at higher leveraging levels and longer time intervals, as should also be expected.

From equation (11), I reduced the number of trades by 10, factored them out, and canceled those 10 winning trades, giving:

$$F(t) = \bar{e} \cdot b_0 \cdot (1 + \gamma)^{N-10} \cdot [(1 + 0.0454)^{10} \cdot (1 + 0.0454)^{393-10} \cdot (1 - 0.0271)^{370}] \quad (12)$$

In the equation, the 10 removed trades were at the same average percent profit per trade as the others. The average loss per losing trade remained the same, meaning we only removed trades that generated a profit.

While we may not have specific details on the trades removed, it's reasonable to assume that the average of those 10 trades would align closely with the overall average. The fact that these 10 or 20 trades, on average, yield a similar average percent profit per trade is not a surprise but rather an expected outcome.

The place for each $\pm r_i$ in the series of returns in equation (8) from $i = 1$ to N does not matter as long as trade i is somewhere in the series before the end at time T , $T + 5$, or $T + 10$.

The same argument holds for every trade. We can still express the series of returns as $(1 + \bar{r})^N \equiv \prod_1^N (1 \pm r_i)$.

These simulations give more of the same at their respective leverage levels. They can serve as reasonable estimates of those future outcomes.

The 5 and 10-year estimates also give us an idea of the potential of the TQQQ strategy.

Even with the removal of some trades, it could generate outstanding performance results exceeding anyone's wildest dreams. Yet, we are only making sensible projections of continuing to do what the strategy had done for the previous 14.4 years. We are indeed playing on the variance of a 3x-leveraged QQQ scenario.

ON ADDING MORE TIME

Adding 5 years to the trading strategy is straightforward: $5 \times 52 = 260$. It would add 260 trades to the 763 executed in the prior 14.4 years. It is pushing the number of trades to 1023. Adding 10 years will increase the number of trades N to 1283.

We know the number of trades to be added to the system before making those 5 or 10-year estimates. Moreover, the average profit per winning trade will further stabilize near its long-term average of 0.0454, the average value the strategy has attained over its first 14.4 years of weekly trading.

Surprisingly, or not surprisingly, the average profit per winning trade ($\bar{r}_+$) will tend to rise slowly over the added years.

The ETF's average long-term volatility would be close to the same average it had during its past 14.4 years. It is expected to continue to fluctuate around its acquired long-term mean. Our trading will not change the nature of the market.

I only state: it will just be more of the same.

We are not introducing anything new to the strategy, only a continuation on the same basis it has operated over the past 14.4 years.

The foreseeable future will likely look like the past. A simple and daily unpredictable continuation of what has been with the same long-term trends and mean-reverting variance as before.

The profit targets won't change; the time limit for any trade will still be 5 trading days. The 7% and 8% profit targets will be the same with no assurance of being hit. But some will, even though we do not know how many or when, just like in the past. However, we expect they will occur close to their acquired long-term averages.

Based on equation (8), the order of the sequence of returns is immaterial. You will get the same answer no matter how that series of returns is sequenced and get the same averages.

The removed trades are at the average percent win per winning trade. Each of those trades could be canceled anywhere within the series of 763 trades. It is not the same as removing the largest gainers; it is more something that does not affect the overall long-term averages. After removing the 10 or 20 positive trades, what will be left will have about the same average percent win per winning trade¹⁵.

The win rate, which was at 51.51% for all non-leveraged scenarios, would go down to 50.86% for the minus 10 trades and to 50.20% for the minus 20 trades. These are almost insignificant moves on the win rate ratio, but they will nonetheless significantly impact the portfolio's outcome.

The TQQQ strategy has unique properties. No matter the leveraging factor, entries, and exits remained the same for all simulations. The reason is simple. The Monday open and the Friday close were the same for all trades. Those were not optional or conditional entries or exits. They occurred no matter what. Also, the sell limit orders at 7% and 8% would be executed at 7% or 8%.

Every single trade was predetermined and would be the same for all simulations, whether over past or future price data. Our trading decisions could not alter the price series.

We might not know the future opening or closing prices, but we have averages on

¹⁵ As illustrated in equation (12).

the percent variations, which is what is being played here within the average volatility percentages.

There are no price predictions made, none at all. The strategy is playing the percentage variation game, not a price or winning amount game.

The Friday's close will be at whatever level of return, positive or negative. The other trades will close with a 7% return or an 8% return. Those will add up and positively impact the average percent per winning trade, which without the profit targets could have been in the vicinity of the average percent loss per losing trade ($\approx +2.71\%$). It's all very simple stuff.

Even if you did 100,000 simulations using the Monte Carlo method without replacements, you would get the same final answer every time. The paths series would change. Most probably, none would be the same, but the outcome would remain the same for all simulations.

Also, none of those paths would lead you to bankruptcy. The ride would undoubtedly be bumpy, no matter which path would be taken. But that should also be expected.

And no big surprise, the portfolio metrics of interest would also be the same for all those simulations, as stated in equation (8).

THE PORTFOLIO PRODUCT EQUATION

Furthermore, the portfolio's product equation almost assures you that it will not go down to zero:

$$F(t) = \bar{e} \cdot b_0 \cdot (1 + \bar{\gamma})^N \cdot \prod_{i=1}^N (1 \pm r_i) \quad (13)$$

The product section of the equation might give you major drawdowns, but the long-term view is still a long-term rising market. Given the future value equation, you want the long-term return to be positive $\bar{g} > 0$.

If you look at the market's historical data, it has been in an uptrend for a couple of centuries. Confirming Mr. Buffett's bet on America and the nation's continued prosperity.

This is why, due to the larger drawdowns than usual, and before attempting to use this trading strategy, you should add protective measures to alleviate the said drawdowns.

The fixed stop-loss approach could be costly, as illustrated in the article series: **One Percent Per Week**. You should look at other procedures for your downside protection.

Equation (13) could hypothetically reach zero if a single $r_i = -1$, meaning the stock market dropped -100% in one week. You should expect with a high degree of certainty that it will not happen anytime soon. We would need a mass extinction event for that to happen, manmade or otherwise.

Nonetheless, you might be unable to escape some of those drawdowns anyway, such as the May 2010 Flash Crash. In the simulations, the position that week was still closed the day after the Flash Crash at the close on Friday. And not during the precipitous 36-minute decline the day before.

The Flash Crash occurred only 4 months after starting the simulation. In real life, it might have raised questions at the time as to the viability of the trading program, but regardless, you had to persevere. That one-day awful decline was not even the worst drawdown over the trading interval.¹⁶

Another point, from my observations – and no surprise – is that the +5 and +10-year projection estimates might be underestimated and not overestimated.

The rationale is that we froze the position growth rate at the leverage level reached over the past 14.4-year data. We then applied them to the future when those growth rates would continue to increase since the average of an exponential function sees its mean move toward higher grounds as the time series increases.

The calculations based on the simulations gave a $\gamma = 0.0096$ for the 1.5x-leverage setting over the first 14.4 years. It would rise to about $\gamma = 0.0142$ with the added 5 years and to $\gamma = 0.0224$ for the 24.4-year scenario. It amplified the point that the average position size would be on an exponential curve, as should have been expected and following equation (13).

In the case of the 20 removed winning trades, we are left with 373 winning and 370 losing trades. That is not even a 1% edge; it gets in at 0.20%. Indistinguishable from total randomness.

That series of returns would pass any randomness test. We are as if flipping a fair coin. There is no skill there.

Yet, our trading procedures and gaming rules totally squashed the MPT expectations by generating long-term returns way beyond anything that might be viewed as some future efficient frontier.

Moreover, the Law of diminishing returns should kick in somewhere and show that the overall return will gradually decrease as you add more time. We have been taught that since the 1970s. And yet, with very elementary trading rules, it is not what you

¹⁶ See Figure #4.

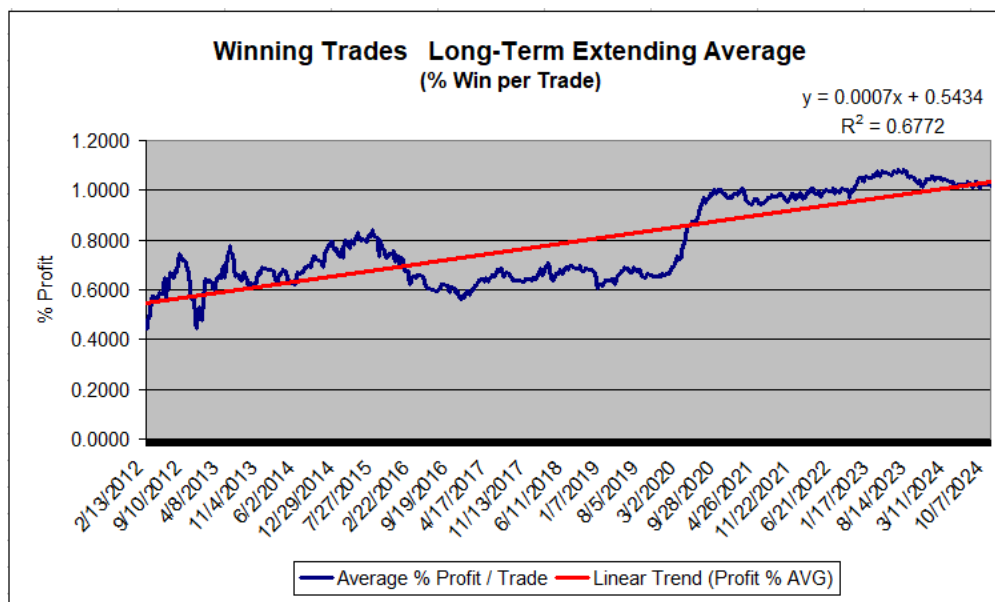
see here.

THE AVERAGE PERCENT WIN PER TRADE

I extracted the expanding average percent win per trade from the simulated data, let the average win rate settle down over the first 104 weeks (2 years), and charted the series from 2012 onward. The percent win per trade increases with time. It is equivalent to the average percent win per week since we only make one trade per week. As illustrated below, it has a positive slope. It demonstrates that what we have learned should be questioned again.

Here is that chart, which spans 12 years, with its first 2 years not shown. What I wanted to see was its long-term average and how it behaved.

Figure #5: Average Percent Win Per Trade



([Click here to enlarge](#))

It's worth noting that the average percent win per trade, as depicted in the chart, is on a positive trajectory over time. The simulation concluded its 14.4-year journey with an average win of 1.0201% per trade, as Figure #2 can attest. We started at a lower point, but the average has consistently risen. The regression line indicates an average increase of 0.000007 per week. To put this in perspective, that is 70 cents on a \$100,000 bet.

It should be considered normal since the amount returned from those trades will increase considerably with time.

Having the weekly return rate increase with time is the same as having your CAGR

increase.

You get an average CWGR, a compounding weekly growth rate that grows marginally over time. The relation is simple: $(1 + \text{CWGR})^{52} = \text{CAGR}$. Since the CWGR is rising with time, so will the CAGR.

Moreover, we can reasonably anticipate that the strategy will continue to exhibit these favorable properties beyond 2024. It suggests that the average percent return per trade will continue to increase, as indicated by the upward trend of the regression line. Such consistent growth over the years will further enhance the appeal of the TQQQ strategy.

It also underscores the point that if you want to reap the benefits of long-term trading, you have to be in it for the long term.

There is also the possibility that after 763 weeks of return data, we still have yet to reach the point where the Law of diminishing returns kicks in. What should kick in, however, are future feasibility constraints. Your portfolio gets so big that it starts to influence the market itself. But you are not there yet; you have a long way to go before that happens. And you have years to prepare for such events.

THE TQQQ STRATEGY'S INCREASING CAGR

The TQQQ trading strategy is a surprising entity. It has very simple trading rules. Yet, it can offer an increasing CAGR over time. How many strategies can do that?

Nonetheless, you could design hundreds of variants of this strategy based on the same theme as the TQQQ strategy. The potential is immense.

Would a fair coin venture that far from its expected mean?

Since the strategy takes a trade every Monday, the minus 20-trade scenario would make 743 bets that the market will rise. It would be correct 373 times and wrong 370 times. That is not a great way to win, with only 3 trades ahead on 743.

Still, you would win the game with a 53.63% CAGR. See Table #2, third line showing a gain of \$326,080,050 for the 19.4-year estimate and over \$3 billion for the 24.4-year scenario.

For the minus 20 trades case, we could adjust equation (12) to give the following:

$$F(t) = \bar{e} \cdot b_0 \cdot (1 + \gamma)^{N-20} \cdot [(1 + 0.0454)^{20} \cdot (1 + 0.0454)^{393-20} \cdot (1 - 0.0271)^{370}] \quad (14)$$

No harder than where 10 winning trades were canceled.

The advantage of the TQQQ strategy resides in its volatility, leverage, and the spread

between the average percent win per winning trade and the average percent loss per losing trade.

It is in the trading methods where I overcompensated for the more than occasional declines. The winning hit rate was at 51.51%, meaning that 48.49% of those trades were losing trades.

From equation (14) we could calculate the edge for a 50/50 proposition: $(1 + 0.0454) \cdot (1 - 0.0271) = 1.01707$. And determine for an even trade distribution as if part of a normal distribution, 370 winning trades and 370 losing trades having for multiplying factor: $(1 + 0.0454)^{370} \cdot (1 - 0.0271)^{370} = 524.516$. This would be confirmed by $(1.01707)^{370} = 524.581$ with the difference due to rounding to the fourth decimal.

All you needed was a compensation factor greater than one to be positive. In this case, we had more than the minimum since $(1 + 0.0454) \cdot (1 - 0.0271)$ is greater than one. Refer to the [Fixed Fraction](#) paper for a more detailed exposé and the formula on setting compensation factors.

Anything done to improve the winning percent spread would benefit the strategy.

THE TQQQ TRADING STRATEGY IS GAMBLING

Don't get me wrong. The TQQQ trading strategy is gambling and playing on the notion that, over the long term, the general market trend is up and that there are more up than down weeks. The strategy will play out and benefit from TQQQ's higher volatility.

The TQQQ strategy will win because it often takes an excess profit percentage within the average inherent ETF volatility range, raising its average percent win per winning trade.

Technically, you are playing volatility excursions, and there are a lot of those. You are also playing the game: "One in the hand is better than two in the bush" gambit. You take that excess weekly profit while it is there and then let it compound week after week after week.

You are okay with what Monday's open price will be. You will wait for a percent change that will execute your limit sell order at its 7% or 8% profit target or take what is left, positive or negative, at Friday's close.

You know you will win the game due to those simple trading rules. It is the only point that should be of some importance: knowing even before you start that you will win the game you constructed.

As mentioned before, the trading rules are so simple you could do it all by hand on

your cellphone from anywhere.

You buy the open on Monday and set a limit sell order at 7% above the opening price. If the price gets near your profit target in the next couple of days, cancel that limit order and reissue a new limit sell order at 8%. If the position is still active by Friday's close, liquidate it and take the closing price, whether at a profit or a loss. Your week is done. Prepare for the following Monday's open.

Overall, only a few minutes a week. Nothing to interfere much with your everyday routine. Your TQQQ program could do the same job automatically in microseconds.

You could also interrupt your trading program for whatever reason you want and at any time you want. You do not initiate a new order or sell your existing position. The program returns to cash whenever it liquidates its position, whether with a profit or at a loss. Remember, you are in charge from A to Z.

You can override the program if you think the market is going down. And wait for your next trade opportunity. You have no obligation to trade. Letting a down week pass you by is a protective measure to add to your trading procedures.

Nonetheless, as shown in Table #2, removing or not executing a might be profitable trade will have some impact, just as not executing a trade that might end up losing. Regardless, you are in charge.

A POWERFUL PORTFOLIO VALUATION EQUATION

Equation (13) was broken down to its constituent factors as in equation (11). The product function $\prod_1^N (1 \pm r_i)$ is equivalent to the separation of the winning and losing trades based on their average percent gain or loss, resulting in:

$$\prod_1^{763} (1 \pm r_i) \equiv [(1 + 0.0454)^{393} \cdot (1 - 0.0271)^{370}] \quad (15)$$

The strategy had 393 trades with an average percent gain of 4.54% per winning trade or as expressed above $(1 + 0.0454)^{393}$. Leaving 370 trades with an average percent loss of $(1 - 0.0271)^{370}$ or -2.71% per losing trade.

Let's define $a_i = (1 \pm r_i)$ or its equivalent $a_i = (1 + \bar{r})$ where $\bar{r}$ has the same average value for all the return rates. We could write $\prod_1^N a_i = a_1 \cdot a_2 \cdot a_3 \cdots a_{N-1} \cdot a_N$ for both definitions, and both would give the same answer. We could also shuffle all the a_i and get the same answer again.

What is important is the sequence of a_i , where each one plays a role in the outcome. If you add trades to the sequence, you will get something like $\prod_1^{N+\nu} a_i = a_1 \cdot a_2 \cdot a_3 \cdots a_{N-1} \cdot a_N \cdots a_{N+\nu-1} \cdot a_{N+\nu}$ as if adding more and more trades, which would

require more and more time since the strategy is structure to only add one trade per week.

The question should be: should it impact the long-term portfolio metrics? Not that much in the strategy's immediate future since it would be minimal, as shown in the change of 0.0000060 in the above example. The average win rate would remain close to its long-term average, oscillating almost randomly up and down around that value.

In the product equation (13), we had nothing to help us determine its value, even though we knew it would stand. However, factoring it into winning and losing trades, we could get a concrete expression translating the strategy's trading behavior and an answer to its value.

In equation (11), we had an expression giving us the portfolio's value after 763 trades spanning 14.4 years; in a trading environment, we have to qualify as almost random-like. Even the hit rate got close to a 50/50 proposition with its 51.51% outcome, which is undistinguishable from a randomly generated fair coin-toss series.

And yet, we got a relatively close representation of what the portfolio did over those 14.4 years of trading one of the most highly volatility ETFs.

Table #3: List Of Variables And Portfolio Metrics

	Description	Variable	WL8 Simulation Metric
	initial bet	b_0	Starting Capital
	average exposure rate	$\bar{e}$	Exposure
	average bet size growth rate	$\bar{\gamma}$	<i>Extracted after the simulation</i>
	average percent win per winning trade	$\bar{r}_+$	Avg Profit %
	average percent loss per losing trade	$\bar{r}_-$	Avg Loss %
	total number of trades	N	Position Count
	number of losing trades	λ	Count of Unprofitable Positions
	number of winning trades	$N - \lambda$	Count of Profitable Positions

All the variables used in equation (11) come from the Wealth-Lab simulation metrics.

Here is a more generalized representation of equations (13) and (11):

$$F(t) = \bar{e} \cdot b_0 \cdot (1 + \bar{\gamma})^N \cdot (1 + \bar{r}_+)^{N-\lambda} \cdot (1 - \bar{r}_-)^{\lambda} \quad (16)$$

Equation (16) can represent the **One Percent Per Week** portfolio. All that is missing is the average bet size growth rate $\bar{\gamma}$. The simulation results are giving us all the metrics to figure it out:

$$\left[\frac{F(t)}{\bar{e} \cdot b_0 \cdot (1 + \bar{r}_+)^{N-\lambda} \cdot (1 - \bar{r}_-)^{\lambda}} \right]^{\frac{1}{N}} - 1 = \bar{\gamma} \quad (17)$$

where we rearranged equation (16) to extract the average bet size growth rate $\bar{\gamma}$, which is expected to be low but still increase as the number of trades increases. Equation (17) has a similar structure as equation (4).

Equation (16) was the one used to get the results in equation (6). We need to account for the exposure, which will directly affect the outcome since it is a scaling factor.

Any trading script simulation, at least using the Wealth-Lab 8 program, will give you all the needed portfolio metrics, as described in Table #3 above.

In the **One Percent Per Week** trading strategy, the number of trades N is predetermined over the trading interval and from the start. You will have 52 trades per year for the number of years the strategy will run.

It is not a guess or an estimate you must make; it is what the program is coded to do: take one trade per week. And the objective is to make that 1% percent per week, as the name of the strategy implies.

You will not get it every week, but it remains the objective, which, on average, is reached.

Equation (16), with a sufficient number of trades to justify generalizations, can help us make reasonable future estimates based on the acquired portfolio metrics.

We can base the future of the trading strategy on its current long-term portfolio metrics on the condition that the metrics are significantly and statistically representative of the portfolio's past. It means you have a sufficiently large number of trades to make reasonable averages.

A POWERFUL EQUATION

To make projections or future estimates, we can add more of the same trades:

$$F(t) = \bar{e} \cdot b_0 \cdot (1 + \bar{\gamma})^{N+\nu+\psi} \cdot (1 + \bar{r}_+)^{N+\nu-\lambda} \cdot (1 - \bar{r}_-)^{\lambda+\psi} \quad (18)$$

where ν is the number of added winning trades at the average percent gain, while ψ is the number of losing trades at the average loss per losing trade. If you add no winning trades, then $\nu = 0$, just as if you add no losing trades, you would get $\psi = 0$. It is getting you back to equation (16). You could also subtract trades from either side to evaluate the impact of those trades.

Furthermore, with a $\bar{\gamma} = 0.0$, it would imply no leveraging and revert equation (18) to equation (6). This $\bar{\gamma}$ factor will come to be interesting.

Equation (18) remains an estimation of what might come. It carries no guarantees. It is only based on a continuation of what it did in the past, which should also be considered a reasonable expectation, especially with those long-term portfolio metric averages.

The percent win to percent loss spread will be important in the above equations. To

be profitable will require an edge: $\bar{r}_+ > |\bar{r}_-|$. Otherwise, you will have to compensate with the number of profitable trades compared to losing trades. This implies having $N + \nu + \psi > \lambda + \psi$, a positive hit rate. You need an edge like in any trading strategy wanting to win the game played: $(1 + \bar{r}_+)^{N-\lambda+\nu} \cdot (1 - \bar{r}_-)^{\lambda+\psi} > 1.0$.

But even more important, you need a number of trades N to be sufficiently large that it is, in fact, statistically significant and representative of what the strategy is doing.

In the **One Percent Per Week** strategy, I used overcompensation on the % win per winning trade to the % loss per losing trade spread¹⁷.

The strategy worked for 14.4 years and gave astounding results.

You want to build on the information gathered in those simulations.

It is as if the simulation results are telling you the story of what is to come. It also tells you which variables would have the most impact if increased or decreased.

Equation (18) is giving you this ability. Sure, the future remains the future, and it is uncertain.

But stating that more Mondays will come your way is not uncertain.

Saying you do not know which way the market will go next week is not uncertain.

Nor should you be astounded that a sell limit order at a 7% profit target is executed if hit.

You play any game, and you will get better at it.

If you practice more than anybody else, you will find yourself at the top.

You would have built all that expertise by being at it for a long time. You can design your trading strategy to do the same and give it habits and procedures that it has to follow because your experience says that the odds in your favor will increase when you do this or that. You even make a system out of those trading rules.

Nonetheless, your interest is anchored on $\bar{g}$ as presented in equation (1). Your goal is to raise $\bar{g}$ to levels exceeding the future general market average. All the equations presented giving $F(t)$ are simply equation (1) variations.

Moreover, equation (18) provides guideline estimates of where the strategy might go. You want an estimate of the portfolio's value in 5 years, adjust the equation, and

¹⁷ See the **Fixed Fraction** paper for ways to structure compensation factors.

add $52 \cdot 5 = 260$ trades to N . Since the hit rate is at 51.51%, you should get close to $\nu = 134$ winning and $\psi = 126$ losing trades.

The market would keep on doing what it has done in the past. Trade after trade, its long-term average metrics would remain relatively stable, having ups and downs, and still stay close to their respective long-term averages.

ADDING MORE TRADES

We saw the impact of removing 10 or 20 trades from the winning trades, but what about adding 10 or 20?

It might justify the quest to find those 10 or 20 additional trades. At the same time, we could also find ways to reduce the number of losing trades with better downside trade management.

To visualize the impact of such moves, I performed the calculations below. It was, again, based on the same TQQQ portfolio metrics used in Table #2. Instead of removing trades, 10 or 20 were added to the winning side of the equation using the same leverage factors and corresponding position growth rates.

Thereby, altering equation (18) as follows:

$$F(t) = \bar{e} \cdot b_0 \cdot (1 + \gamma)^{N+10} \cdot [(1 + 0.0454)^{10} \cdot (1 + 0.0454)^{393} \cdot (1 - 0.0271)^{370}] \quad (19)$$

where I added 10 winning trades at the same average percent win per winning trade as in the other examples.

Table #4 was built on the same basis as Table #2. The two tables consider the impact of having ± 10 and ± 20 trades to the strategy's 763 trades over the 5 and 10 added years while following the same average portfolio metrics.

Where do you find those 10 or 20 added trades?

We should first look at the timeframe over which to find those trades. It could be over the entire trading interval where you could add trades, that is, over the length of equation (19), that it be over the first 14.4, or during the added 5 years (up to 19.4), or over the next 5 years up to 24.4 years. It should give you more than enough time to find something original and consequential.

Those trades could even be added during the last 5 years of your project, just as in the first five. It would still honor equation (19) above.

However, in real life, it would only be in the added 5 and 10 years where the strategy was extended. You can only impact the future. Nonetheless, you could backtest your methods from the beginning to show that they were worthwhile, at least in the past,

and that you could expect the strategy to behave similarly in the future.

For the added 10-trade case, it would raise the win rate to 52.13%, while adding 20 trades would raise it to 52.75%. Either moves not that far from the baseline win rate of 51.51%, as in Figure #2.

Table #4. Five And Ten-year Projections With And Without Added Trades

Leverage	Outcome (\$)	CAGR (%)	+Trades	Est. (T)	CAGR Est,	Est. T+5 years	Est. CAGR	Est. T+10 years	Est. CAGR
1.00	72,792,124	56.94	0	72,815,879	58.04	792,454,150	58.84	8,624,267,020	59.32
1.00	72,792,124	56.94	+10 Wins	113,514,424	62.98	1,235,375,835	62.52	13,444,577,287	62.25
1.00	72,792,124	56.94	+20 Wins	176,960,367	68.09	1,925,857,102	66.28	20,959,074,899	65.23
1.20	175,016,437	67.96	0	175,016,437	67.96	2,568,066,882	68.77	37,681,989,273	69.25
1.20	175,016,437	67.96	+10 Wins	275,991,250	73.36	4,049,699,559	64.85	23,895,567,385	66.12
1.20	175,016,437	67.96	+20 Wins	435,222,950	78.93	6,386,152,414	76.88	93,705,864,305	75.68
1.50	536,131,700	81.54	0	536,131,700	81.54	11,520,572,504	82.34	247,557,812,318	82.82
1.50	536,131,700	81.54	+10 Wins	857,946,249	87.56	18,435,828,298	86.82	396,155,079,961	86.38
1.50	536,131,700	81.54	+20 Wins	1,372,930,880	93.79	29,501,985,678	91.40	633,948,272,159	90.00

([Click here to enlarge](#))

When a position is taken, the exposure is 100%. However, since the actual exposure is close to 50%¹⁸, on average, it means nearly half the trading days are spent in cash. The average winning position is held on average for 3.03 days (see Figure #2). That too, is a long-term average.

It is easily explainable: the strategy will execute its profit target whenever it is hit. And this is any trading day of the week. It does not have to wait for Friday's close. If all trades closed on Friday, exposure would effectively average near 100%.

Finding ways to use that available cash while not in a position would be an exploration venue. How about extending the average holding period per position to something like 70% to 80%? That too, would have an impact.

Look again at Figure #4, where the exposure increased to 77.75%, not that we programmed for it, but because we leveraged the portfolio and where more of the portfolio's cash was used.

THE STAKES ARE HIGH

Tables #2 and #4 were all based on the TQQQ strategy starting with \$100k. These tables show tremendous results.

Those results allow anyone to accelerate their quest for financial freedom as if it is a matter of choice. It is like specifying your degree of trade aggressiveness where you select at what rate your $\bar{g}$ in equation (1) will be.

¹⁸ Figure #2 has it at 51.44%.

My point is that anyone with the means and determination can do this at whichever level that best suits their purpose. Understandably, if you do not have the capital or fortitude needed to carry out this type of trading strategy, I'm sorry to say I cannot help you with that.

If you find leveraging at 1.5x too bumpy or too steep, reduce it to something more to your liking. Or add more protective downside measures to alleviate the impact of drawdowns. Combine your TQQQ strategy with other less volatile scenarios; such a move will reduce the overall volatility.

If you have the time before retiring, do add additional years. They can have a tremendous impact, as illustrated in Figure #4.

You could also add more trading years after retiring. You would be financially independent anyway and could withdraw whatever you need or want from your ongoing and still-growing portfolio.

Equation (2) could be your reference point to continue building your investment fund.

From Tables #2 and #4 at the 1.5x leverage, there is a difference of \$1,163,570,595 between the plus 20 and minus 20 trades. The difference in win rates is relatively small; we go from 50.20% for the minus 20 trades to 52.75% for the added 20 trades scenario. However, the difference in outcome is considerable. The portfolio value with the added ten years for the two scenarios has a difference of \$537,276,551,245 after those 24.4 years.

You will probably not get that far, but anything close would be more than acceptable. In 10 to 15 years, your positions would start to impact market prices. You would need other procedures to alleviate or spread those high-volume trades. However, you have 10 to 15 years to solve those problems, and there are solutions, too.

The standard deviation after 763 trades based on the portfolio's hit rate is expected at 13.8, so adding or subtracting 10 or 20 trades is in line with expectations. The ± 20 trades would be within 1.45σ as if the building of Tables #2 and #4 were just ordinary estimates and within $\pm 2\sigma$, giving you a wide range of possibilities.

You could run the TQQQ strategy for 10 to 15 years, which would be enough to get you started. But, it is not the assets of TQQQ that should make the mark; it is the assets held by QQQ.

TQQQ's current total assets are in the order of \$25 billion, which will continue to grow. However, that is not the same as the QQQ ETF, where actual shares are held. TQQQ trades derivatives, swaps, futures, and options to keep their 3x-leveraged pledge.

It also gives you 10 to 15 years to design other programs with similar long-term

potential or better. There are over \$15 trillion dollars in assets in the shares of the top 10 stocks part of QQQ. It should be enough to satisfy your immediate appetite.

Even in the worst scenario, the minus 20 trades at the 1.5x leverage factor would be more than acceptable at any level, whether in 14.4, 19.4, or 24.4 years. Considering the impact of adding another five years, estimates would go through the roof. You could even have 30-plus years while in retirement to keep your portfolio growing.¹⁹

You only have so many years before retirement, so your solutions might be limited. You could continue since the strategy will also work fine while in retirement.

I have made many suggestions in that direction in recent articles²⁰.

Here are the things that could help you accelerate the process; refer to equation (18).

- increase your initial capital b_0
- increase average growth rate $\bar{g}$
- increase time interval t
- increase the total number of trades
 - increase the number of winning trades
 - decrease the number of losing trades
- increase the leveraging factor $\bar{\gamma}$
- find other ways to raise your portfolio's CAGR
 - add new positive trading procedures
- add more money as you go along
- extend the strategy beyond your retirement date
- or any combination of the above

The above suggestions get you right back to equation (11) with the added leveraging function that you can amplify or not. To accelerate building your long-term investment fund, you can select at what level and speed you want that portfolio to grow.

As structured, the **One Percent Per Week** trading strategy plays an exponential game, or gambles if you want, when every Monday at the open, it bets its accumulating cash portfolio long b_{i-1} .

¹⁹ Refer to equation (2) for more detail.

²⁰ See in particular the **One Percent Per Week** article series.

QQQ ON STEROIDS

Leveraging TQQQ is putting QQQ on steroids. From a 3x-leveraged QQQ scenario, you can push it up close to a 6x-leveraged gambit and only pay the fees of a 2x-leveraged scenario. It's no wonder your long-term CAGR can fly.

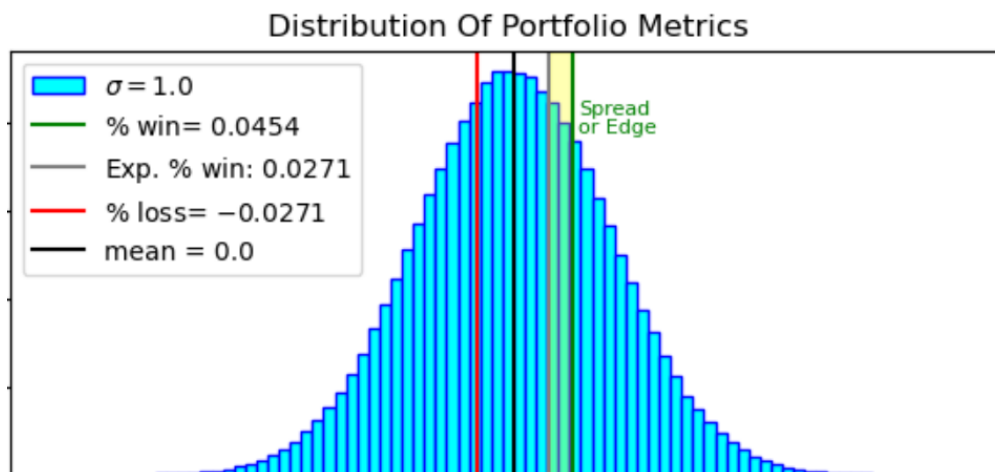
The distribution of the percent average winning and losing trades should have been about the same near $\pm 2.71\%$, respectively. This assumption is easy to make.

Negative trades had at most 5 trading days and could not decline by more than the weekly decline variations. And since the returns are almost normally distributed, as Figure #6 illustrates, our positive side of the equation, without our gaming interference, would also be near the $+2.71\%$ mark. Just like for the downside where no action was taken to minimize or interfere with negative weeks.

With remarkably simple trading procedures, we raised the percent win per winning trade, giving us an edge, as shown in the chart below. The strategy increased the average percent win per winning trade to $+4.54\%$.

The game is definitely about $\bar{g}$ as in equation (1); we are just looking at variations of it. The rest is about what you can do within the time constraints and capital available. You need to take control. What will become the most important are your trading methods and the perseverance you put into carrying out your program.

Figure #6: TQQQ Strategy - Portfolio Metrics



([Click here to enlarge](#))

The TQQQ trading procedures displaced $\bar{r}_+$ (the green line) to the right in Figure #6 (the percent win per winning trade). It does not seem like much, but it made all the difference. And it will continue to do so.

ON THE RANDOMNESS OF PRICE SERIES

It is like when you look at a stochastic representation of prices like a Geometric Brownian Motion (GBM): $\frac{dS}{S} = \mu dt + \sigma dW$. We could substitute μdt for $(1 + \bar{g})^t$ and σdW with a residual of mean zero, almost normally distributed as shown in the above chart (Figure #6).

We accept a GBM equation's stochastic nature but do not seem to like the consequences: the σdW part of it tends to a mean close to zero and a quasi-normal distribution, at least near the mean.

In this strategy, we did not look at the impact of fat tails since positions were limited to 5 trading days. On average, only so much volatility can occur in one week, giving a cutoff point to the distribution.

Nonetheless, it did not stop the strategy from having a -19% drop in one week. Meaning that even if it does not consider fat tails, it could suffer from their occurrence once in a while. Regardless, the system recuperates and goes even higher than before, reminding us that the market's general trend is still up.

The σdW part of the equation puts you right back into a gambling mode where the randomness of the outcome $\pm\%$ of each bet b_{i-1} could be amplified by σ_i , which is also a randomly evolving factor.

You are left to play with μdt . But on a log scale, μdt or $(1 + \bar{g})^t$ are superimposed straight lines. You can grab any segments of those lines, but they will remain straight.

What will bother you is the random side of the equation σdW even if you played the straight line. Over the short term, you still might not know the outcome of the stochastic equation since μdt could easily be drowned by the high degree of random noise generated by σdW .

The noise could account for over 95% of the price move. How could you determine any signal's value in all that noise? Would you even see a positive μdt value in all that noise during a price decline?

THE LONG-TERM RETURN DEGRADATION

In Figure #6, the return distribution shows that for a 5% rise, there is also a -5% decline. They cancel each other out and can, over the years, almost destroy your portfolio's return.

The outcome of this is: $(1 + 0.05) \cdot (1 - 0.05) = 0.9975$ and not 1.0. But when this happens 100 times, we get $((1 + 0.05) \cdot (1 - 0.05))^{100} = 0.7785$.

The same thing will happen for any return level you want to consider. For each positive return, there is its negative counterpart. The more trades you make, the more pronounced the decline in value, just because your trades' percent returns are quasi-randomly distributed. This is by no fault of your own; it is just how the market behaves, with its almost unpredictable ups and downs.

It gets worse if you add more trades: $((1 + 0.05) \cdot (1 - 0.05))^{200} = 0.6061$, but would get much worse if you consider higher return rates like $\pm 10\%$ or $\pm 15\%$. In such cases, you would get: $((1 + 0.10) \cdot (1 - 0.10))^{200} = 0.1339$ and $((1 + 0.15) \cdot (1 - 0.15))^{200} = 0.0105$.

It is like designing a trading strategy that has a 10% or 15% profit target with a stop-loss of -10% or -15%. People say they would not do that, but I see a lot of trading strategies still doing it. Over the long term, such a strategy will have an uphill battle to maintain a positive CAGR, if any.

You have to compensate for that inherent return degradation. Again, refer to the **Fixed Fraction** paper for ways to structure your compensation procedures. The equations for doing it are also provided. It is very easy to do, and while at it, go for the overcompensation procedures just as I did in the **One Percent Per Week** strategy.

If you have limited time, perhaps already 50 years old, there's still a way to significantly boost your CAGR for the remaining 15 years before retirement. The TQQQ strategy could be a game-changer, accelerating the outcome of equation (1). Also, consider increasing your initial stake for even greater potential returns.

Even though you have tools to increase your portfolio return, you should remain in control. It is not a program that should be responsible for your portfolio's outcome. You decide to let it run or not.

Regardless, do add some protective measures, even if it is just going to the sidelines, and let what you might consider bad times pass you by. The game is for the long term. A skipped week here and there is not that important, and it is better to miss a bad down week than have to recuperate from one.

In the end, it will be your steadfast determination to carry it all out that will ultimately matter the most.

You know you stand on solid grounds and that the principles you adopted have a mathematical and solid foundation.

If QQQ can easily have a long-term 15% growth rate²¹, you can imagine that going for up to 6 times the leverage should get you close to a 90% compounded return

²¹ Refer to **QQQ To The Rescue** for a rundown on trading QQQ.

which is what was demonstrated in these pages.

The TQQQ strategy is doable by anyone. It is always your choice to do it or not.

MORE ON THE LEVERAGING GROWTH RATE

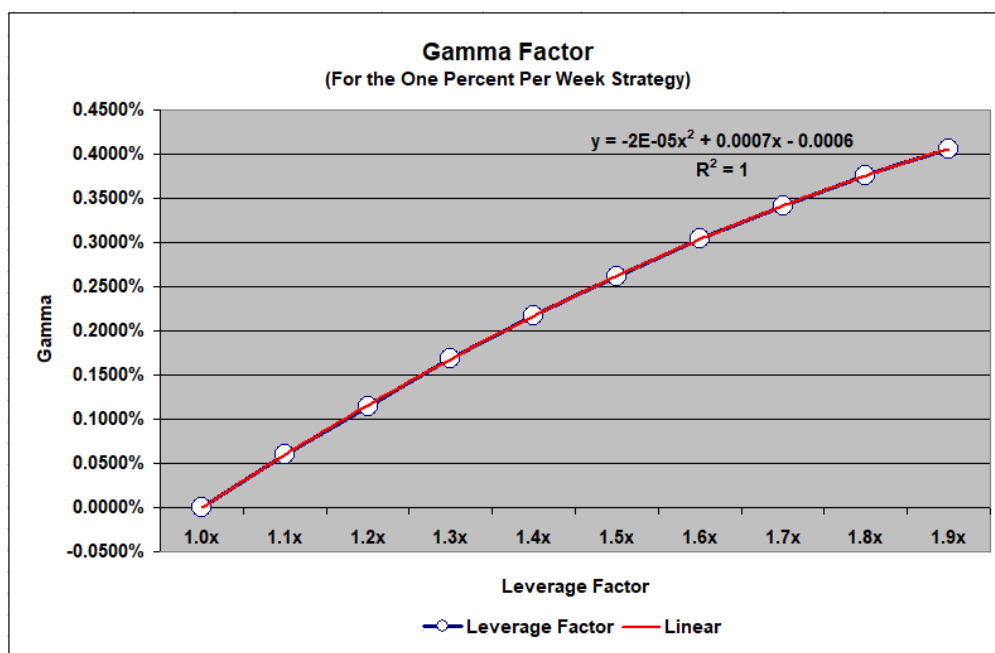
We have a leveraging function $(1 + \bar{\gamma})^N$ embedded in equation (16). Its purpose is to account for the increasing position size over the trading interval should leverage be applied.

With an average $\bar{\gamma} = 0$, we have the no-leverage scenario as shown on line #1 of Tables #2 and #4, respectively. Both come from the portfolio metrics given in Figures #2 and #4, which corroborate those simulation numbers.

Equation (18) is more elaborate. It confirms and verifies all the trading scenarios in Tables #2 and #4, including their forward 5 and 10-year estimates. It also makes provisions to estimate adding or removing winning or losing trades. It is an equation tightly related to the degree of leverage applied on the 3x-leveraged TQQQ ETF.

That we have an equation that explains all those results (in Tables #2 and #4) is in itself a validation of the equation. Otherwise, the results of all those simulations would not match the result of the equation. The equation is there to confirm the simulation result. Or the simulations confirm equation (18) as a worthwhile representation of the portfolio outcome.

Figure #7: The Relation Between $\bar{\gamma}$ And Leverage



[\(Click here to enlarge\)](#)

A correlation between $\bar{\gamma}$ and the leveraging factor should exist. It might be best illustrated with a chart (Figure #7).

In Figure #7, there is a blue line underneath the red line, the regression line over the gamma factors. The correlation coefficient is 100% ($R^2 = 1.00$), stating a perfect positive match. Also, confirming that with zero leverage applied, the gamma factor is at zero. At zero, the gamma rate of change would not affect equations (16) or (18). And again, validating the results of the first line in Table #2 and the portfolio metrics in Figure #4.

The above chart also confirms that for any leveraging value, we can have a gamma factor that responds to equation (18). Note that in Table #1, the percent profit per trade remained the same (1.02% per trade or week) at all leverage levels, as should have been expected since we are only playing percentages.

The leverage and the gamma function are increasing the bet size, not the percent profit. It means that all this does not affect the price series itself. It is the hand dealt where you intentionally choose to increase or decrease the position size.

I would like to entertain using the gamma factor $(1 + \bar{\gamma})^N$ to drive the portfolio's outcome. As you would raise the gamma factor, it would insert leverage in your game. It would allow you to predetermine how your trading strategy will behave even when faced with uncertainty.

It might sound strange at first, but with the gamma factor, you could modulate your market participation and adjust your bet size to what you find acceptable as you go along. You might want to implement it in real-time. It would allow you to change your stance and the degree of trade aggressiveness. Also, providing you with an added degree of flexibility in your game.

So, from equation (18), we have in the same order as in the equation:

- $\bar{e}$, the average exposure rate we get after a simulation
 - For the no leverage scenario, it would remain the same
 - The exposure should increase as we raise the leveraging factor
- b_0 , the initial bet is the initial capital put into play
 - The bet size will increase in line with the gamma growth function
 - The bet size will be exponential and in step with the leveraging factor
- The gamma factor increases the bet size at the specified rate over the interval
 - Either as leverage driven or deliberately set as a position sizing function
- Number of winning trades at their average percent win per trade

- Number of losing trades at their average percent loss per trade

In the simulations, as shown in Table #1, the distribution of trades and their average percent win and percent loss did not change with the leveraging factor. They remained the same as if describing the inner workings of the strategy.

The strategy would make $N - \lambda$ positive trades and λ negative trades at their respective percent gain or loss. This is understandable since the strategy is to play on return averages.

IMPACTING THE STRATEGY FURTHER – OR UPPING THE ANTE

One of the things not sufficiently elaborated about equation (18) is the ability to increase the percent win per winning trade or reduce the percent loss per losing trade.

The impact could be considerable since it applies to the compounding rate of return over the entire trading interval.

It would require some program modifications to expand the yellow band in Figure #7, moving the green or red line further to the right or shifting both to the right simultaneously.

We could transform equation (18) into:

$$F(t) = \bar{e} \cdot b_0 \cdot (1 + \bar{\gamma})^{N+\nu+\psi} \cdot (1 + \bar{r}_+ \cdot (1 + \zeta))^{N+\nu-\lambda} \cdot (1 - \bar{r}_- \cdot (1 - \kappa))^{\lambda+\psi} \quad (20)$$

where $\zeta > 0.0$ increases the average percent win per winning trade, and $\kappa < 0.0$ would reduce the percent loss per losing trade.

Using equation (9) with its 1.5x-leveraged scenario and adding a 10% increase to the average win rate per winning trade while keeping the same number of trades would give:

$$0.50 \cdot \$100,000 \cdot (1 + 0.00262)^{763} \cdot (1 + 0.0454 \cdot \underline{1.10})^{393} \cdot (1 - 0.0271)^{370} = \$2,943,688,797 \quad (21)$$

compared to the \$536,273,572 in Table #1 and Figure #4.

It makes the quest to find that 10% increase per winning trade more than worthwhile.

The prize is over \$2 billion for finding ways to slightly improve on the **One Percent Per Week** program objective. The impact would be even more pronounced with the added 5 or 10 years of trading.

Based on the same scenario as above, adding 5 years of trading would result in \$112,368,810,358, while adding 10 years could generate \$4,289,430,850,649 after those 24.4 years of trading.

The numbers might look astronomical, but if we accepted equation (9) as a reasonable representation of the **One Percent Per Week** strategy, then that conclusion was forthcoming.

Another observation is that we might still be underestimating the strategy's capabilities.

However, you could only get such numbers if you found ways to increase the average percent win per winning trade by those percentages.

These calculations were made to show how worthwhile a quest to find those higher average returns might be. Note that my program version has not yet reached those levels either.

Even a slight improvement, like increasing the average percent win per winning trade by 5%, would be appreciated:

$$0.50 \cdot \$100,000 \cdot (1 + 0.00262)^{763} \cdot (1 + 0.0454 \cdot \underline{1.05})^{393} \cdot (1 - 0.0271)^{370} = \$1,257,425,686 \quad (22)$$

and again, bringing interesting results. The added 5-year scenario would raise the outcome to \$36,024,315,328. While the added 10 years would push those results to \$1,032,069,974,149. All within the realm of possibilities. But there are still 24.4 years to get there or about. And the need to find that added 5% increase on the average percent win per winning trade.

A NUDGE

Improving the percent win per winning trade by 5% will increase the 0.0454 in equation (22) to 0.0477. It's not what you would call a big move by any measure.

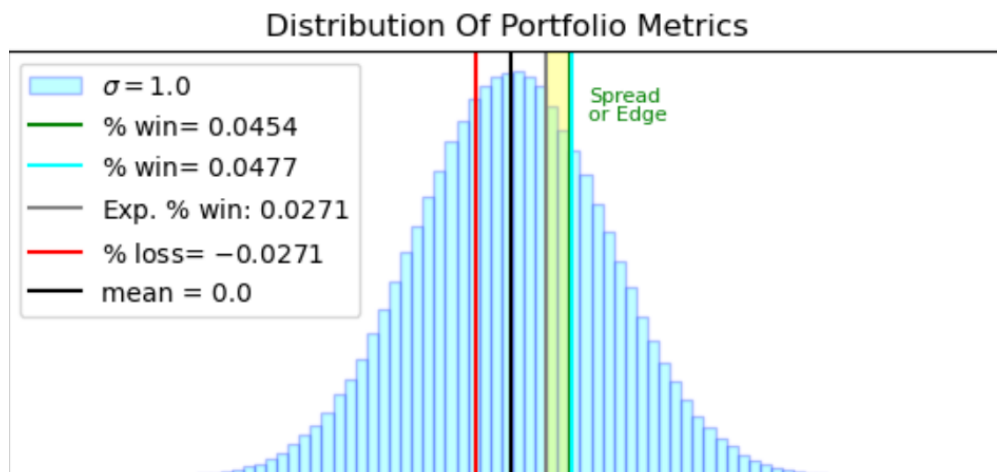
For more emphasis on the above point, I added to Figure #7 the impact of raising by 5% the average percent win per winning trade, giving us Figure #8.

The impact is barely noticeable (see cyan line at 0.0477). Any improvement you can make on the percent win per winning trade can considerably impact your strategy's performance.

As said before, there is always a better trading strategy out there. And you have equation (20) that can guide you to T, T+5, T+10, or more. You could even make adjustments for the years in between.

What the strategy is requesting is your capital, determination, and perseverance.

Figure #8: TQQQ Strategy - Portfolio Metrics - Win Rate on Winning Trades



([Click here to enlarge](#))

Note that adding a zero to the initial capital would multiply all those result numbers by 10. This is another way to stress the importance of your initial capital in this compounding game.

If you have plans to continue investing while in retirement, look back at equation (2) and what it implies. This TQQQ strategy is so easy to implement you could even do it all on your cellphone in less than 10 minutes a week using buy and sell limit orders.

A WALK FORWARD

The last program modification was in May 2024, about six months before writing this.

Since then, the portfolio metrics have been relatively stable, as should have been expected.

The average percent profit per trade stayed close to 1.02% per trade over those 6 months. The same went for the average percent win per winning trade and the average percent loss per losing trade.

As should also have been expected, the number of trades increased: both the number of winning trades and the number of losing trades. We added 24 trades (24 weeks) since the last modifications to the program (Table #5, line 4, mod 2)²².

We can consider those 24 weeks as a walk forward since the program code has been frozen in time since May 2024.

Those 24 weeks are all part of a future the strategy had never seen and could not

²² The modifications are described in the first two articles of the series **One Percent Per Week**.

foresee, especially with the simple trading instructions of this strategy. The stock market went its way, and the strategy executed its program code, whatever was thrown at it.

After those added 24 weeks, all the portfolio metrics we followed stayed relatively close to their acquired long-term averages from last May. The CAGR remained about the same within a $\pm 1\%$ range. The same goes for the percent profit per trade, the average percent win per winning trade, and the average percent loss per losing trade.

Table #5: The One Percent Per Week 6-Month Walk Forward²³

Line Note	Outcome (\$)	CAGR (%)	% Per Trade	Win %	Loss %	# Wins	# Losses	Extracted From
1	410,001	6.93	0.23	1.44	-1.38	460	345	Part - I Using QQQ
2	10,681,520	39.05	0.86	2.73	-2.44	392	222	Part - I TQQQ
3	41,484,956	52.89	0.97	4.68	-2.93	380	361	Part - I mod 1 - May
4	59,176,233	56.66	1.02	4.54	-2.68	380	362	Part - II mod 2 - May
5	70,799,226	57.58	1.03	4.53	-2.65	386	367	Stupendous - July
6	72,792,124	56.94	1.02	4.54	-2.71	393	370	Early Oct
7	74,081,532	56.90	1.02	4.52	-2.70	395	371	Sim. Oct 14

As the program added trades, week after week, it was expected to execute trades in the same fashion as it did over the past 14.4 years. These long-term averages will evolve slowly since any new trade will have less and less influence on the average, whether it be the average percent win per trade or the overall CAGR.

Even after 6 months, the average percent win per trade remained remarkably stable. We might not know the future, but we know how our trading strategies are structured to execute trades.

Usually, when we have a walk forward, strategies break down or start slowing down. Look at [The In and Out Stock Trading Strategy](#) for instance, which showed incredible potential with an 83.5% CAGR over 13.24 years.

The outcome of its future, the equivalent of a walk forward, was shown in the following article [A Walk-forward Example](#). It showed the precipitous collapse of that strategy over the next few months. Refer to the first chart in that article where the drawdown was some -85%.

While not all walk-forwards will lead to a collapse, monitoring our strategies for such events remains critical. Designing our trading strategies to avoid such outcomes in our portfolios is more prudent.

Understanding the sustainability of a trading strategy, or whether it exploits some

²³ All the simulations were performed with no leverage, meaning a leveraging factor of 1.0x.

specific circumstances, should be analyzed. We should aim to build strategies of a general nature capable of withstanding future market uncertainties. But maybe most of all, these strategies survive and thrive above market averages for extended periods; read 20 years and more.

IT IS YOUR GAME AND YOUR RULES

Ultimately, you are responsible for your trading strategy, whether you designed it or not. You are the one who allows it to operate. You cannot come back and say: "it is not my fault. I did not understand what was at play when all you had to do was put it to the test and verify that those trading methods were worth your attention".

Do not undertake such an adventure without knowing what to expect; do your homework. You do not have years to waste on trivial pursuits.

A 6-month walk forward is not that much, but on the other hand, we cannot wait 2 or 3 years for more evidence of the merits of our trading strategies. It is preferable to add protective measures in case things go wrong.

The worst thing to happen is having your portfolio go down in value, knowing you could have bypassed the problem with a bit of coding. On the other hand, you should not panic because you had a down week here and there; no matter what you do, you will have those, too.

In any real-life stock investment scenario, you are always at the right edge of your stock charts, facing an uncertain future. It is up to you to make reasonable assumptions of what could or might happen. Having redundant protective code for the just-in-case bad moments is better than not having them at all.

So, design the needed protection to alleviate or at least minimize those downfalls.

OPPORTUNITY COSTS

The opportunity cost of waiting a few years might appear as not that high to gain more clarity on our trading strategies, but that would be a misconception. It is not the initial delay that is that important in an exponential function; it is the value of the last few years that will matter most; case in point: the last doubling time.

If, on average, your doubling time is about three years, that delay would also move your endpoint by three years. It could cost you the last doubling time in the series, which could be very expensive, in the tens and tens of millions.

You have such a series on page 17. If you get to the 12th doubling time, you should get $2^{12} \cdot F_0 = 4096 \cdot F_0$. The next one would get you to $2^{13} \cdot F_0 = 8192 \cdot F_0$.

Missing the last doubling time is equivalent to throwing away all you have earned over the previous 12 doubling times (over $3 \times 12 = 36$ years). And losing another one would put you back to $2^{11} \cdot F_0 = 2048 \cdot F_0$, reducing your potential portfolio by -75% .

For example, removing from Mr. Buffett's personal portfolio his last doubling time could represent more than $-\$50$ billion, whereas removing his first doubling time would have been about $-\$10$ million.

The opportunity cost of the last doubling time could be staggering.

Somewhere along the line, you have to decide: am I doing this or not?

That is the question. So, make sure you answer that question to your satisfaction since, in the end, it is all about you and what you can do.

THE RESTART FROM DAY ONE

You will have to restart the program from zero, meaning from day one. Continuing where the program ended would require a lot of capital and might be prohibitive. The 5 and 10-year extensions were to show how the program would behave going forward; the same for the added and removed trades to give a probable range of the outcome.

You should also expect something similar as your strategy evolves over its first 15 years, on its way to 20 and then 25 years. You will get something close to the same picture as in Figures #1 and #3, with their respective scales. Figures #2 and #4 are there to corroborate those charts.

After a restart from day one, you should expect your chart to slowly evolve into Figures #1 or #3 or something in between or better. That is your part. If you go fully invested in SPY, QQQ, or TQQQ, you will find that you have, every Monday, a bet on the future with your money on the line.

You might discuss the volatility of each ETF while all three are simultaneously moving in the same direction at their respective long-term growth rates ($\bar{g}$). Nonetheless, whichever ETF you may choose, it will still be your money on the table, and this, week after week, for years on end.

What looks like a flat section of the equity curve in Figure #1 is the build-up, as needed for an exponential function. As your portfolio increases in value, the 7% or 8% percent profit amounts grow accordingly. What was at first a \$7,000 profit could, in the later stage of the chart, represent \$7,000,000 and more. The stakes get higher and higher with time, and so does your bet size.

What you have is not just a simulation over past data; it could also be your scenario

starting from scratch. And as said often, there will be more Mondays, more Fridays, and more price fluctuations going forward. It is up to you to profit from all that volatility to the best of your abilities.

The TQQQ trading strategy is gambling. It is easy to agree on that.

Every Monday, it will bet that the market will go up by the following Friday. If your profit targets are hit, all the better; you take the money and run to come back the following Monday.

The thing is, the way you structured the game with your trading rules, you have extracted a positive edge from this erratic and almost randomly evolving market ecosystem.

You turned the casino rules around, putting the edge on your side; you are now the House. So, take advantage of it and buy your financial freedom.

ADDENDUM

Follow the link for my copy of the Wealth-Lab 8 program: [One Percent Per Week](#). All the simulations presented in this document used that program version with my minor modifications. Dion (aka Glitch) is the original program designer. My appreciation for a job well done.

There is still what I often referred to as a procedural "bug" in the code, but it is more of a program feature. I treated it as such, mostly because I have yet to work on it. Its effect is that it chickens out of trades if it sees a bit of red and chooses to get out at break-even. It often turned out to be the best course of action. The reason I kept it. Nonetheless, my next task is to circumvent it and give it a more optimistic view and more resolve in the face of adversity.

A lot of variations on this theme could also be highly profitable, many even more so.

Related Papers and Articles:

[High Stock Portfolio Returns? Easy](#)

[Your Stock Trading Portfolio Destroyer](#)

[You Will Earn Every Penny You Make](#)

[Stock Trading Strategy Alpha Generation](#)

[There Is Always A Better Retirement Fund: \[Part I\]\(#\), and \[Part II\]\(#\)](#)

[Welcome To YOUR Stupendous Retirement Fund](#)

[The One Percent a Week Stock Trading Program: \[Part VII\]\(#\), and \[Part VIII\]\(#\)](#)

[The One Percent a Week Stock Trading Program: \[Part V\]\(#\), and \[Part VI\]\(#\)](#)

[The One Percent a Week Stock Trading Program: \[Part III\]\(#\), and \[Part IV\]\(#\)](#)

[The One Percent a Week Stock Trading Program: \[Part I\]\(#\), and \[Part II\]\(#\)](#)

[The Long-Term Stock Trading Problem: \[Part I\]\(#\), and \[Part II\]\(#\)](#)

[Anticipating A Stock Portfolio's Long-Term Outcome](#)

[The Big Open Project](#)

[Sitting On Your Bunnies Might Be Your Best Investment Yet](#)

[Self-Managed Retirement Funds](#)

[Make Yourself A Glorious Retirement Fund](#)

[The Age Of The Individual Investor](#)

[QQQ To The Rescue](#)

[Take the Money and Keep it – II](#)

[Use QQQ - Make the Money and Keep IT](#)